Airfoil rollout30

GT, EigenFluid K1024, Laplacian K1024, Neural Kinematic Bases K10

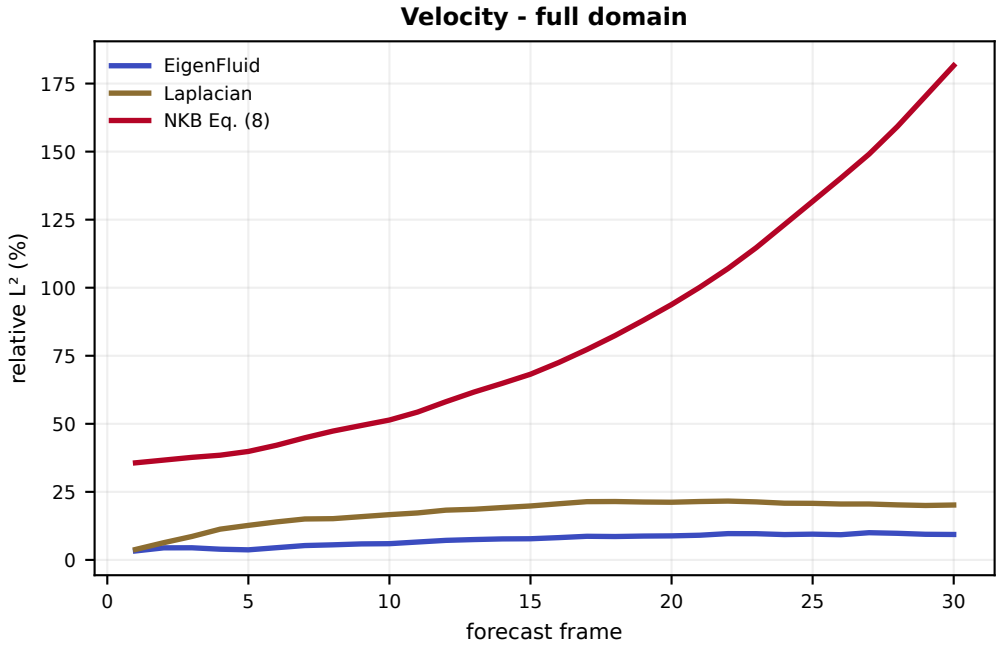
Mean / final relative L2 error (%)

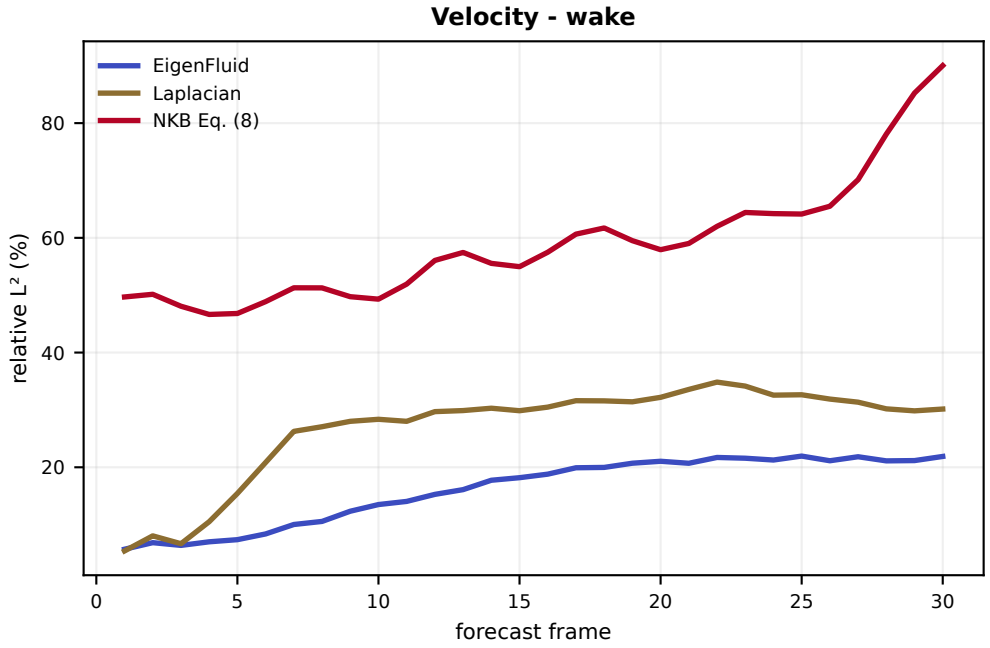
Method	velocity full	velocity wake	vorticity full	vorticity wake
EigenFluid	7.39 / 9.35	16.14 / 21.89	54.17 / 64.51	78.58 / 98.25
Laplacian	17.53 / 20.17	26.76 / 30.15	91.85 / 97.38	100.76 / 107.55
NKB Eq. (8)	84.07 / 181.61	58.93 / 90.03	120.92 / 151.95	111.89 / 134.05

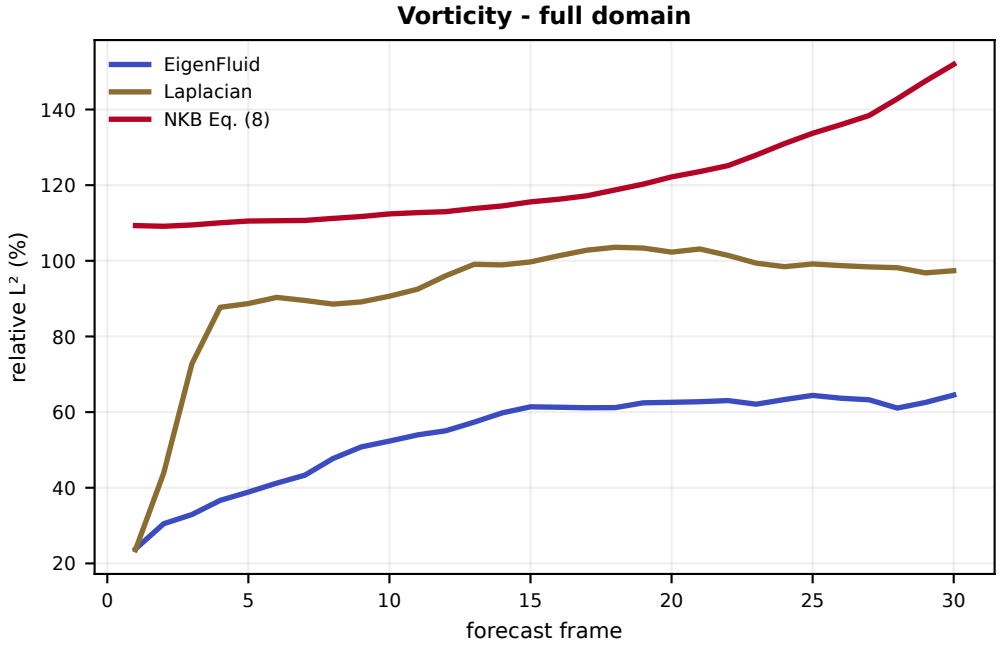
Method	Propagation
GT	D2Q9 TRT-LBM Navier-Stokes
EigenFluid	pure-vorticity epoch 5000; variational ROM
Laplacian	same variational ROM; rectangular basis
NKB	camera-ready Eq. (8) reproduction

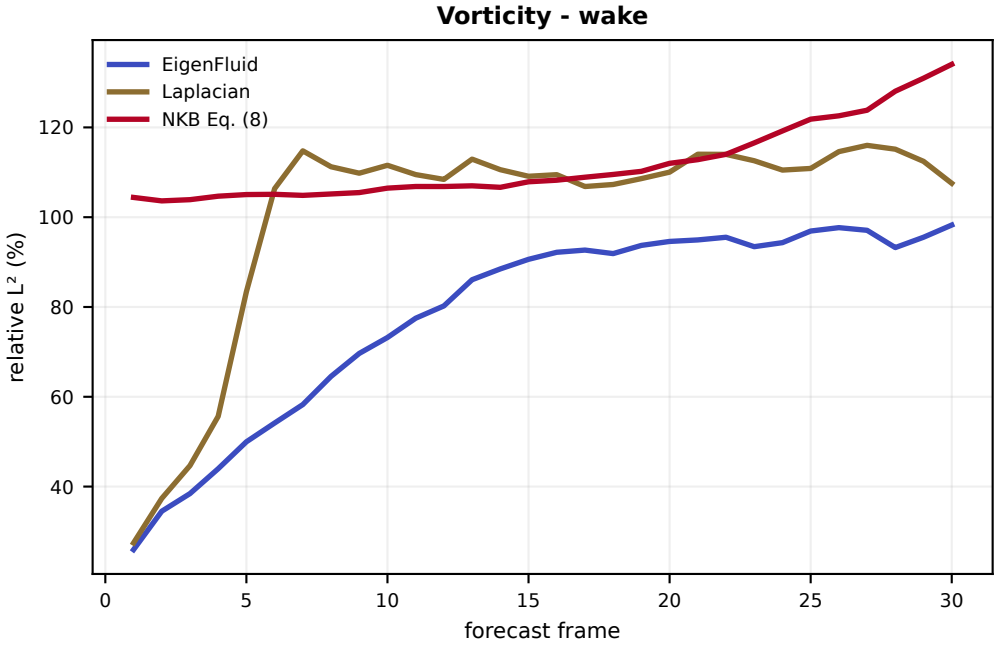
Rollout error curves

Relative L2 against the same GT trajectory



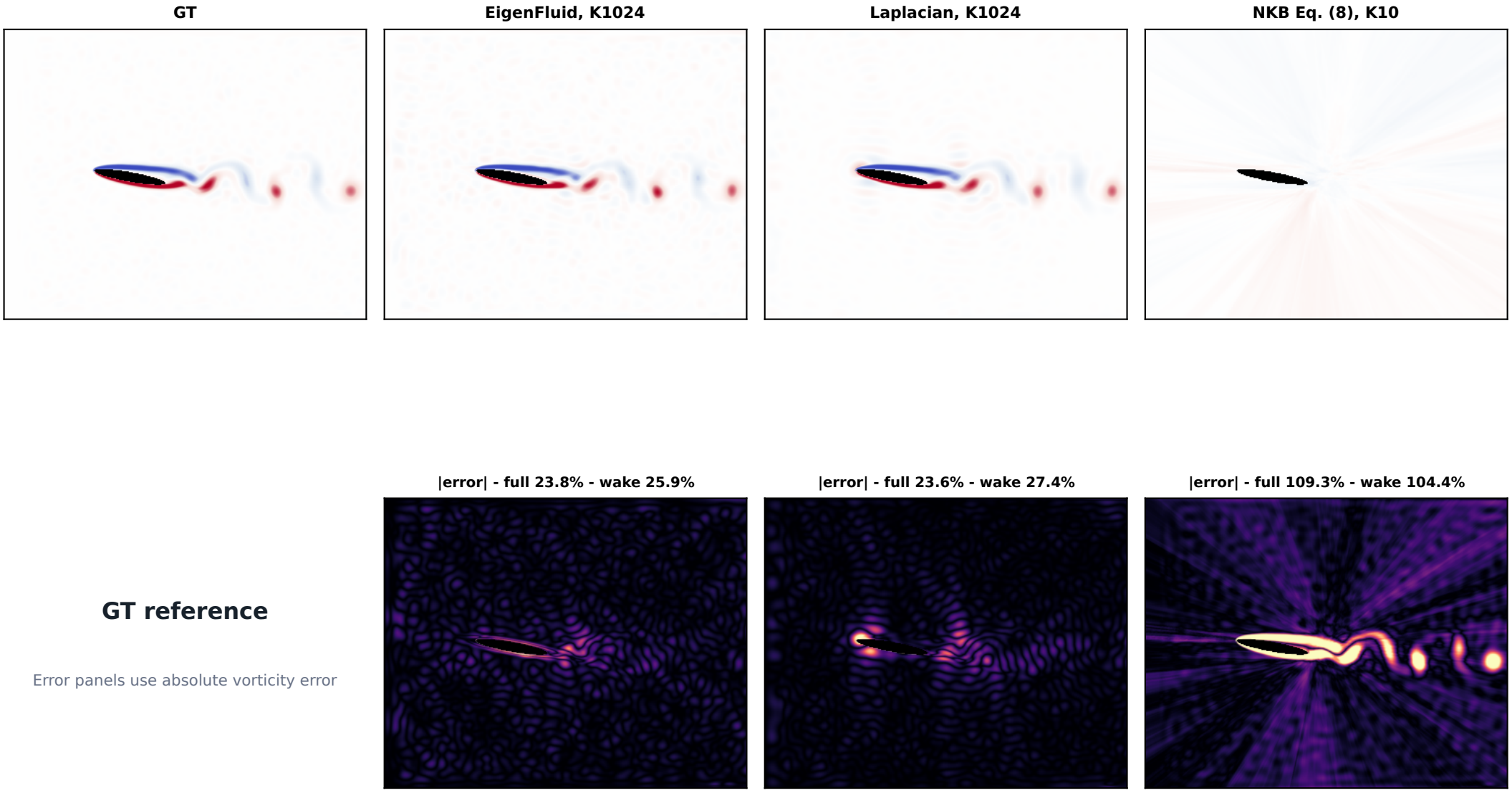






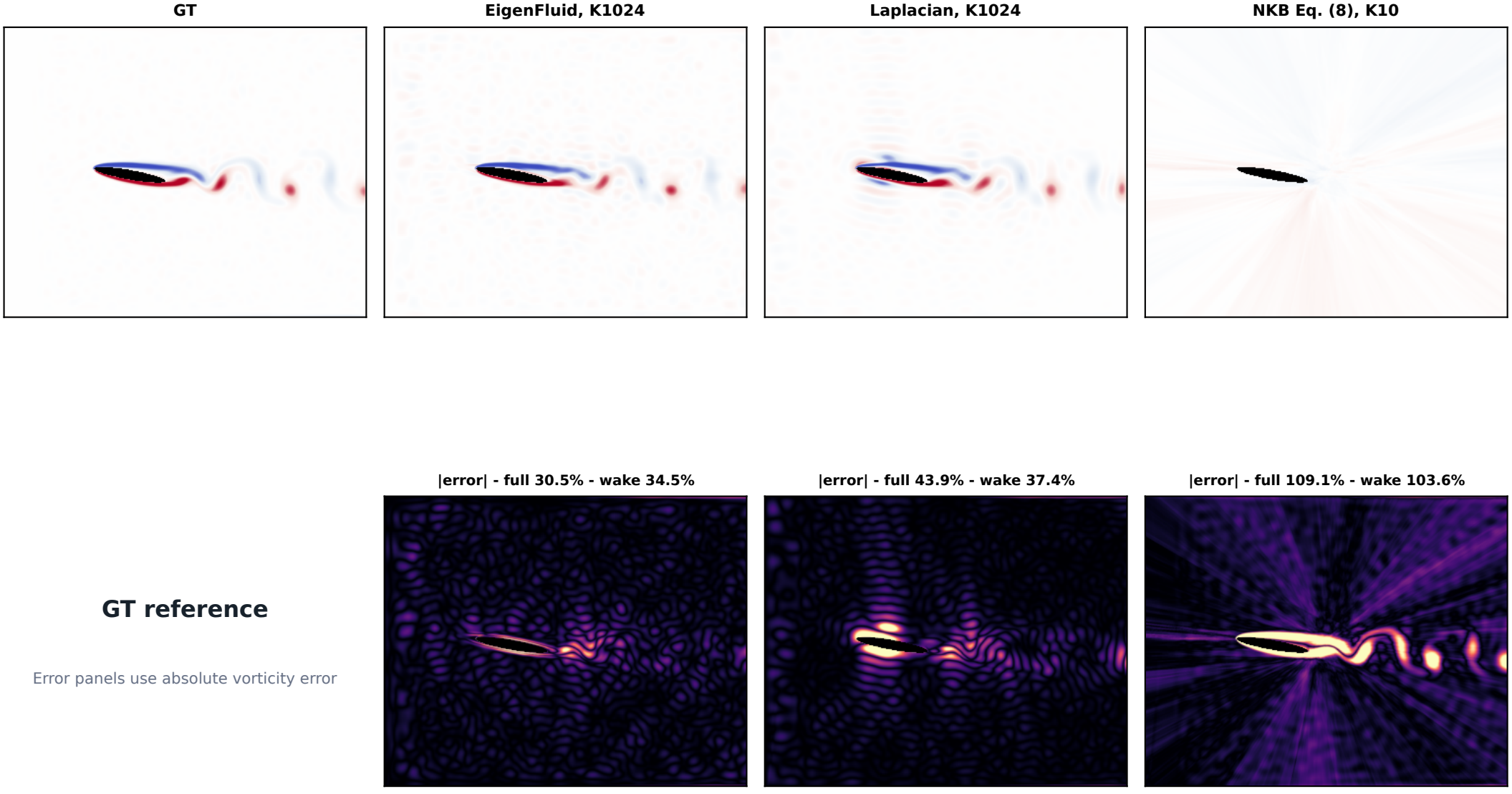
Vorticity rollout - frame 01/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



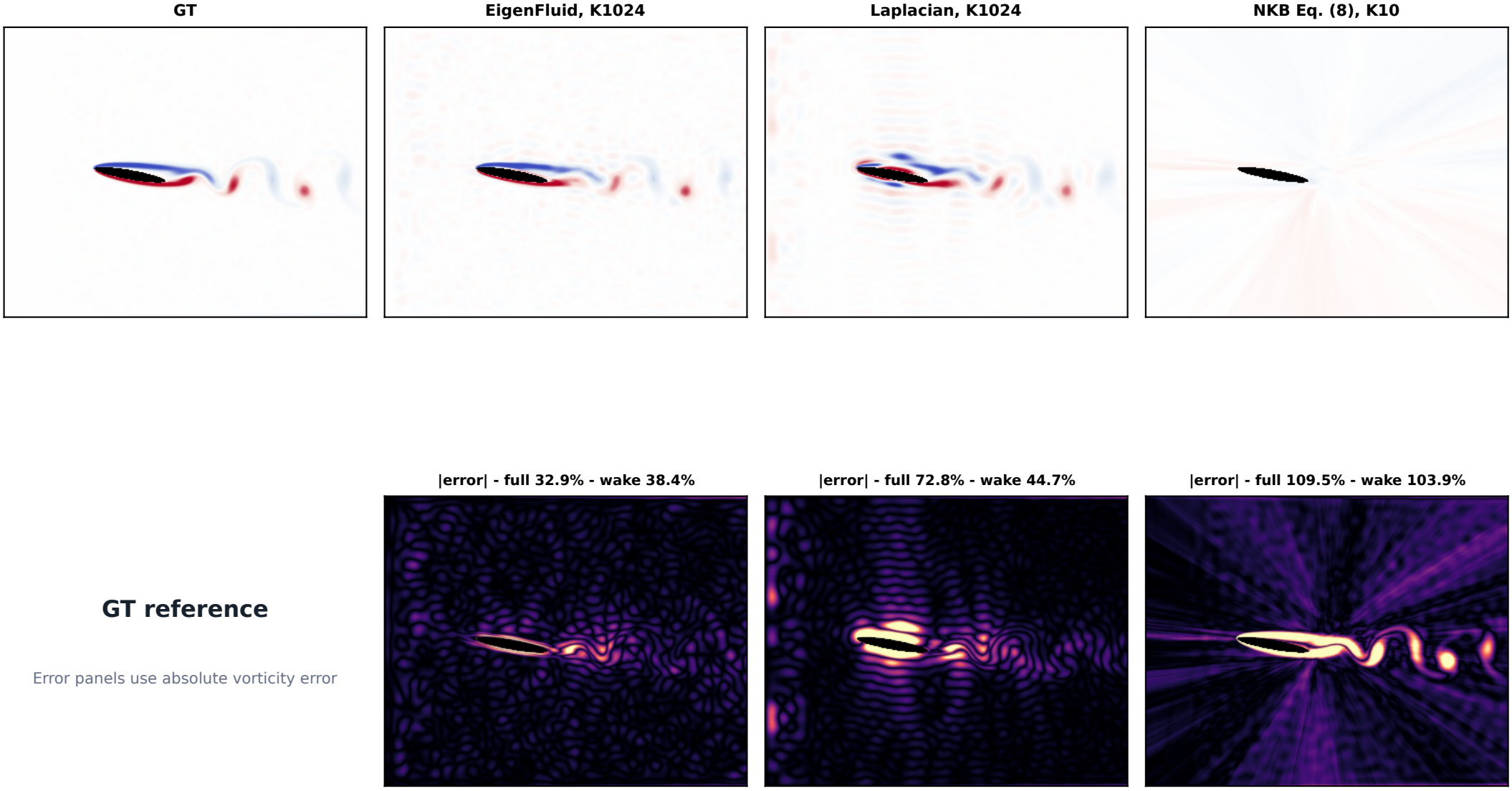
Vorticity rollout - frame 02/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



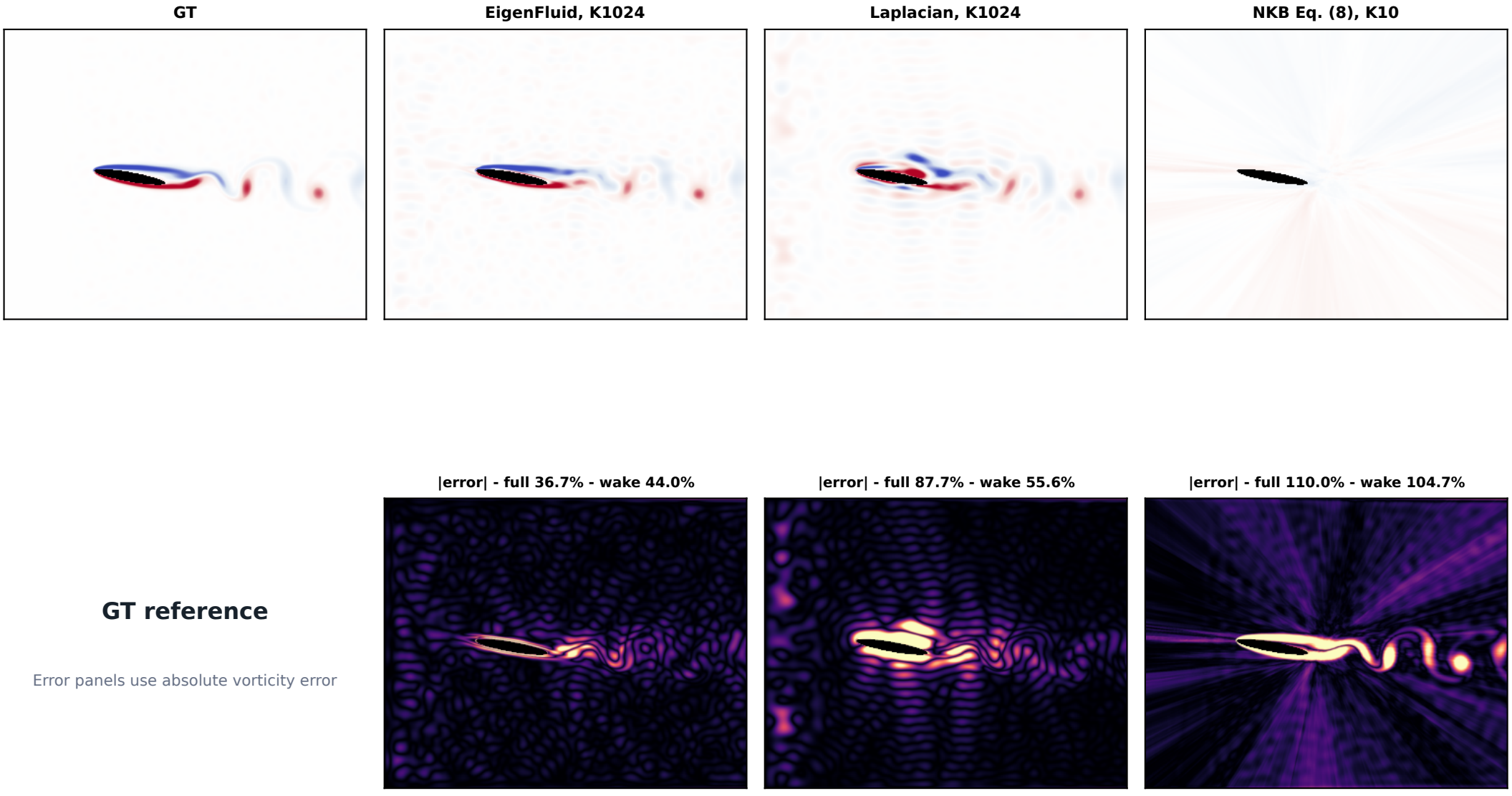
Vorticity rollout - frame 03/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



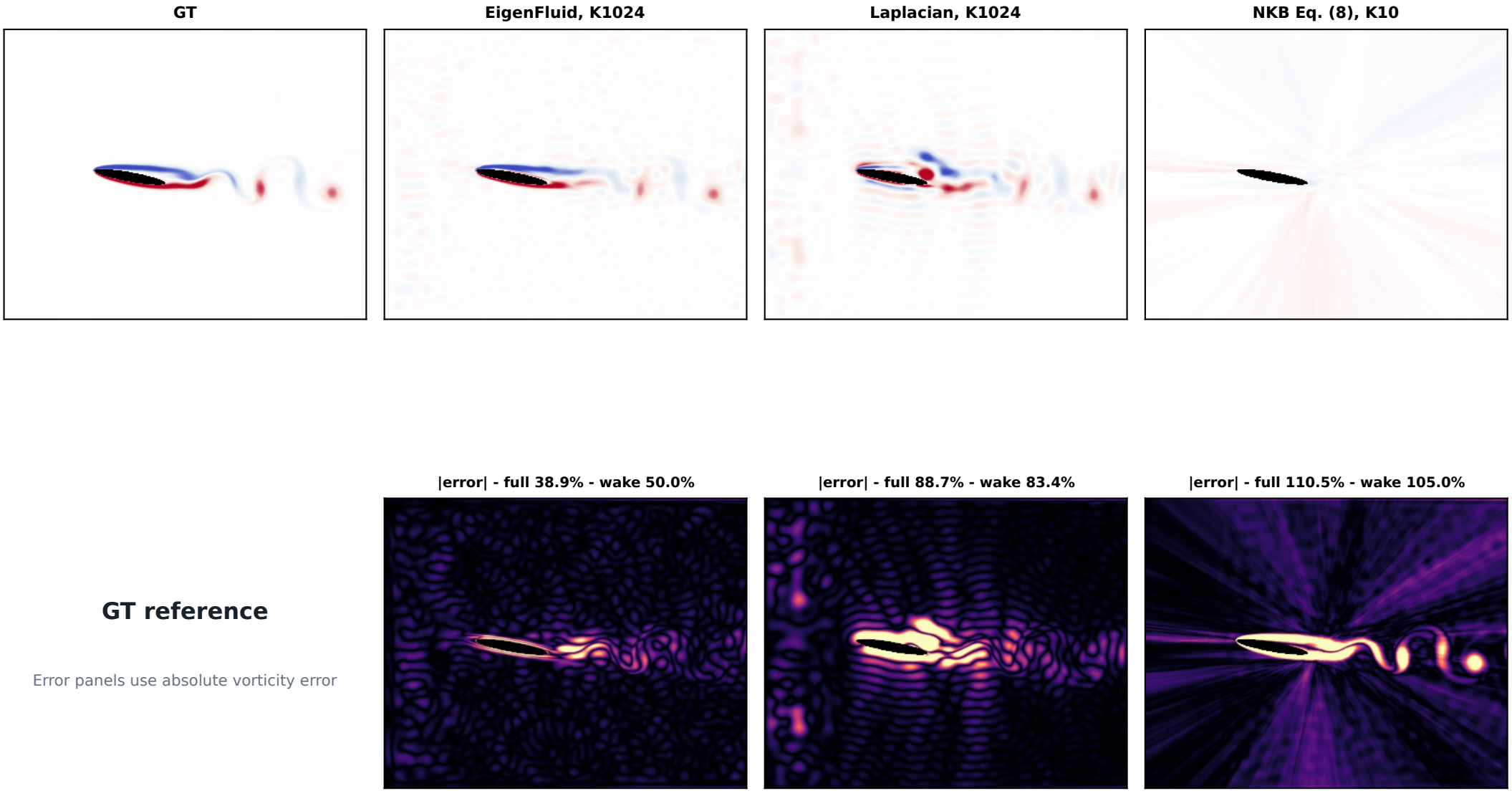
Vorticity rollout - frame 04/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



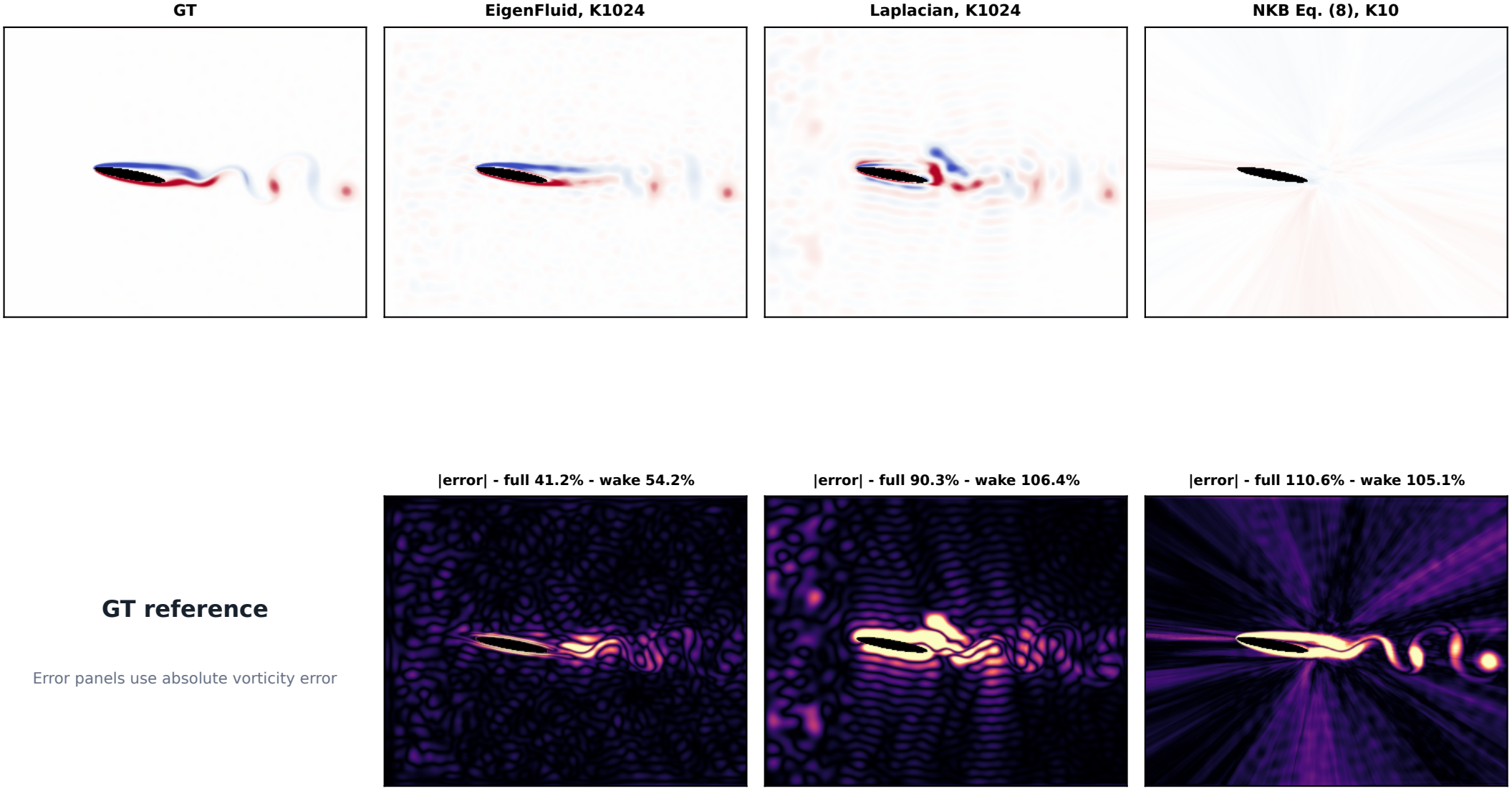
Vorticity rollout - frame 05/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



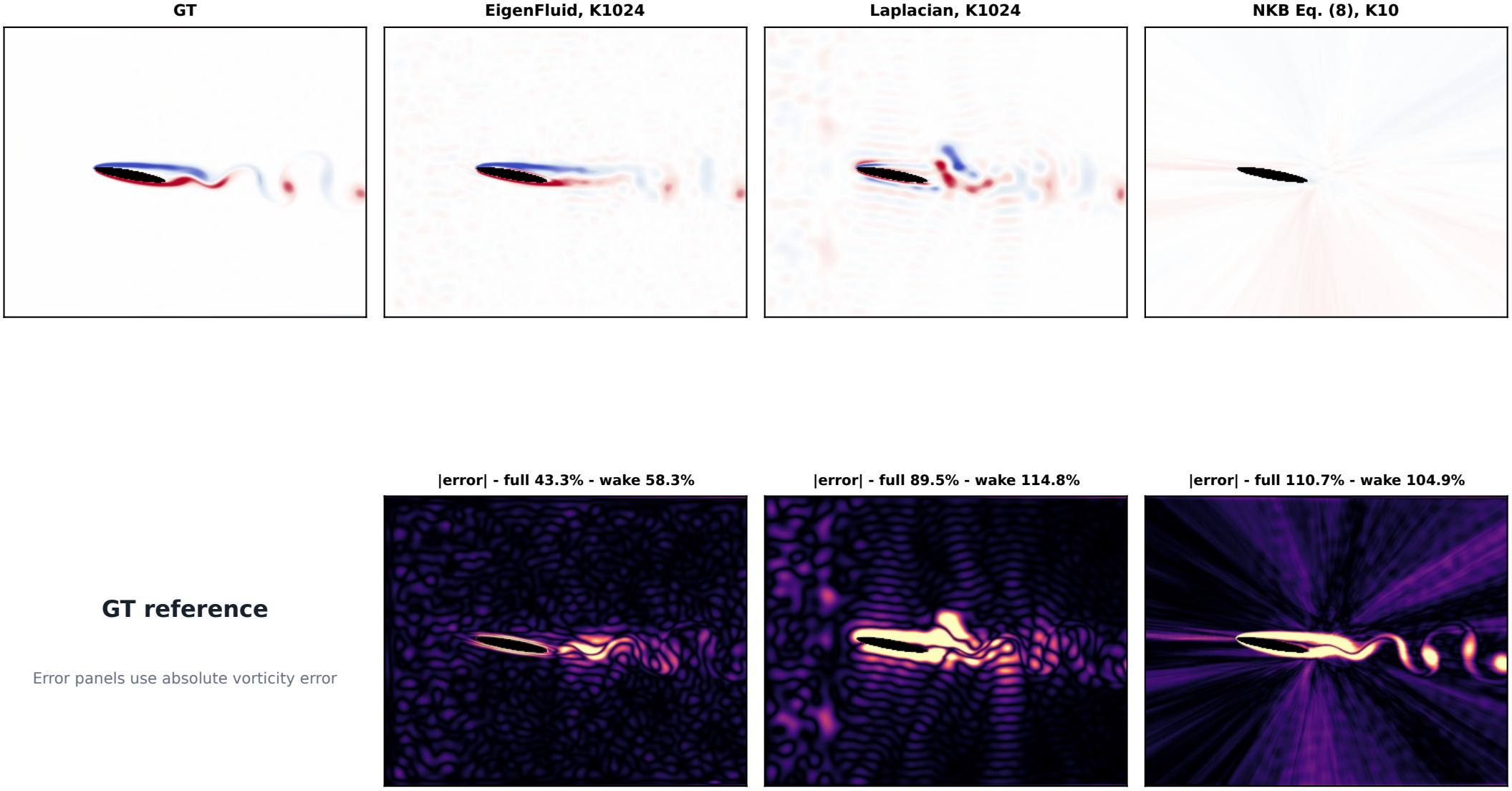
Vorticity rollout - frame 06/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



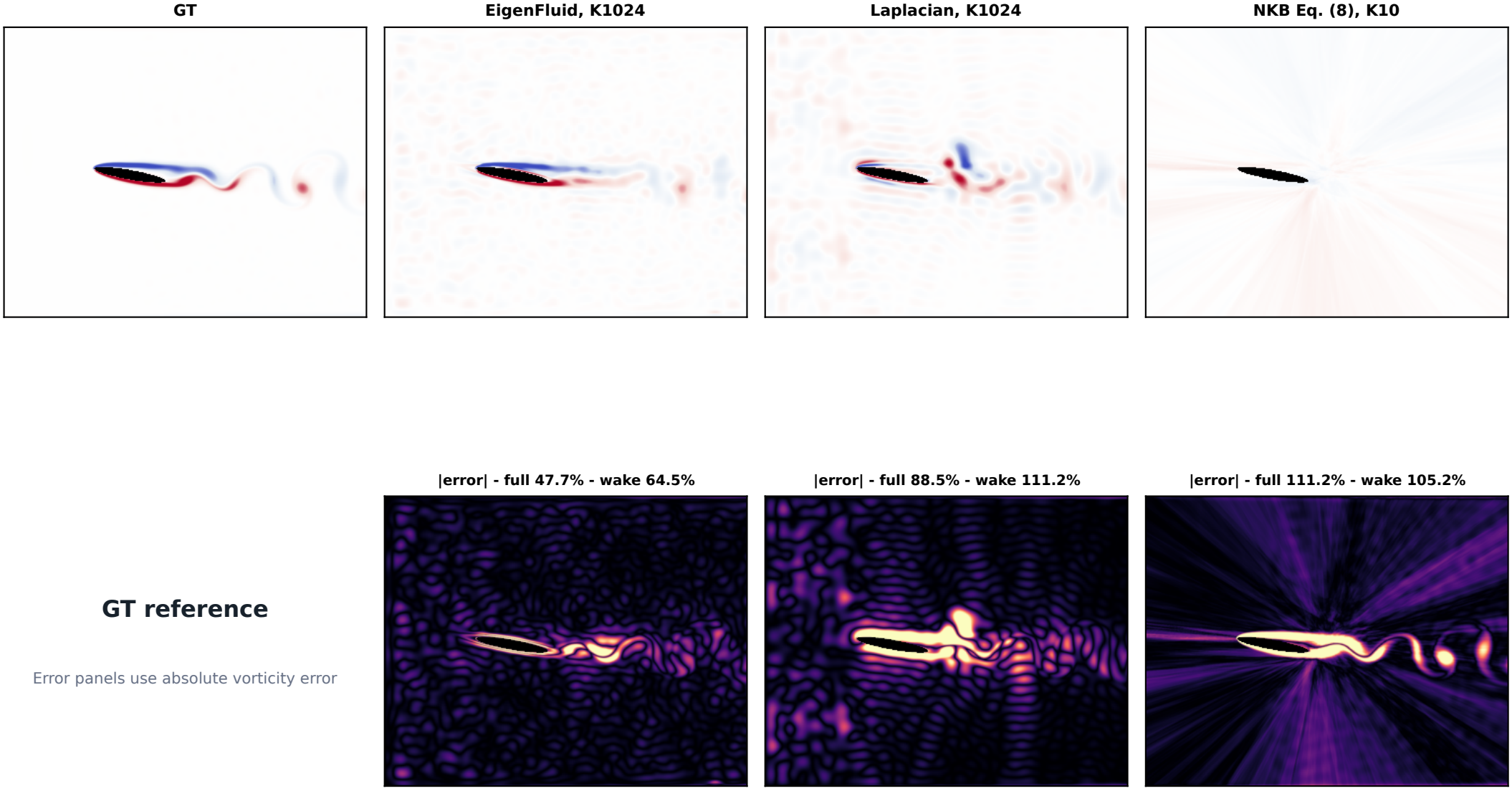
Vorticity rollout - frame 07/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



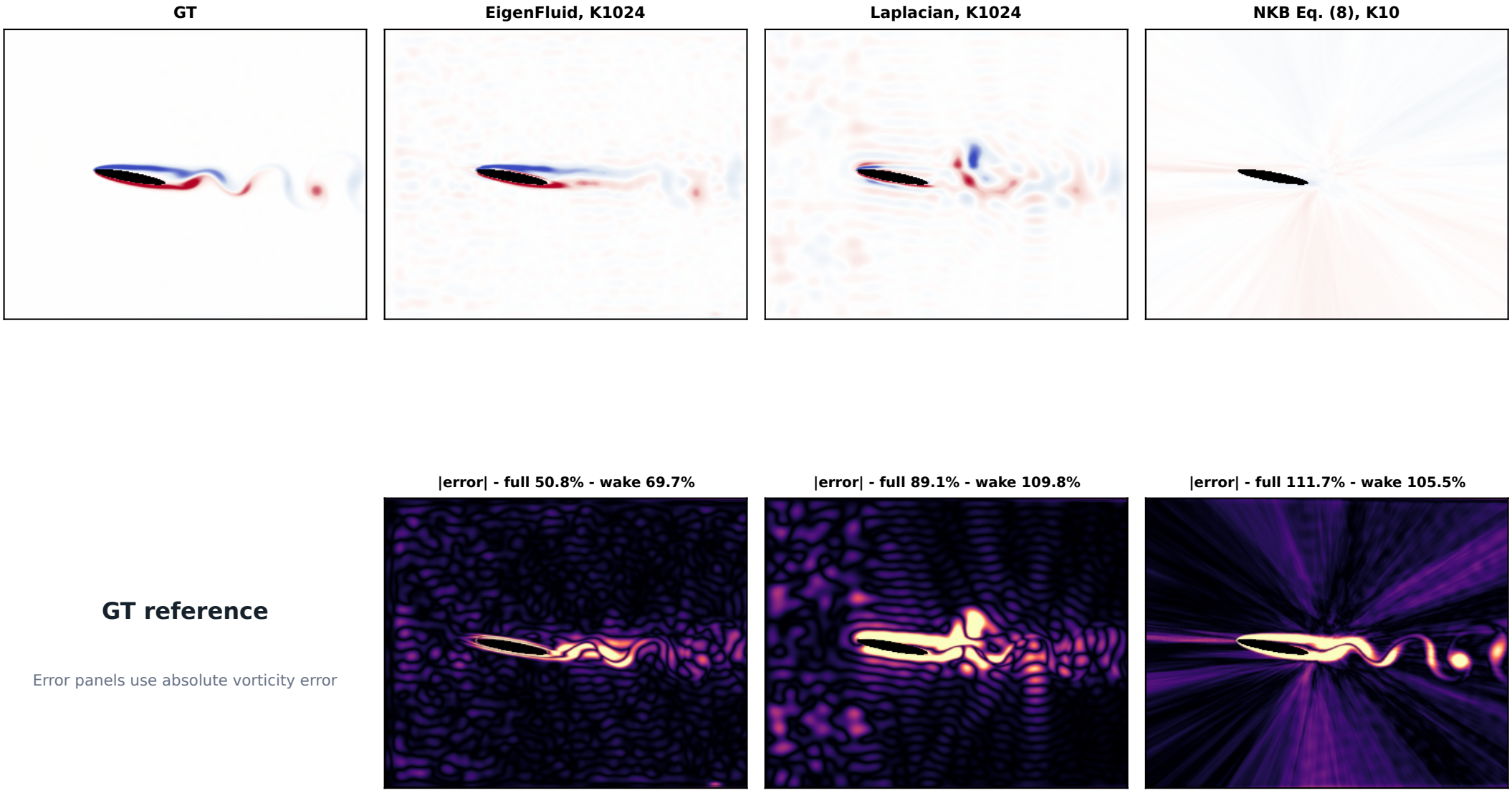
Vorticity rollout - frame 08/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



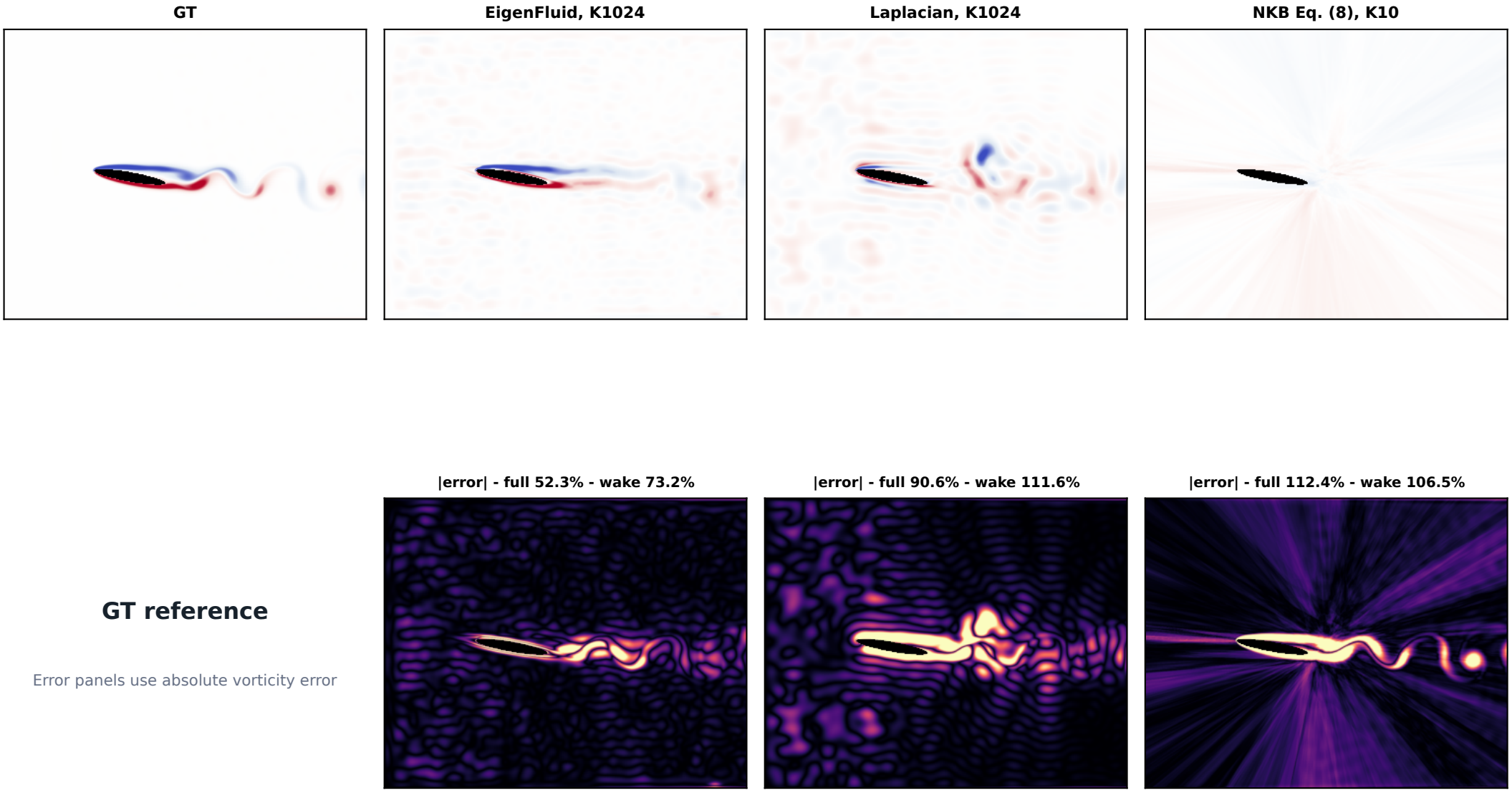
Vorticity rollout - frame 09/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



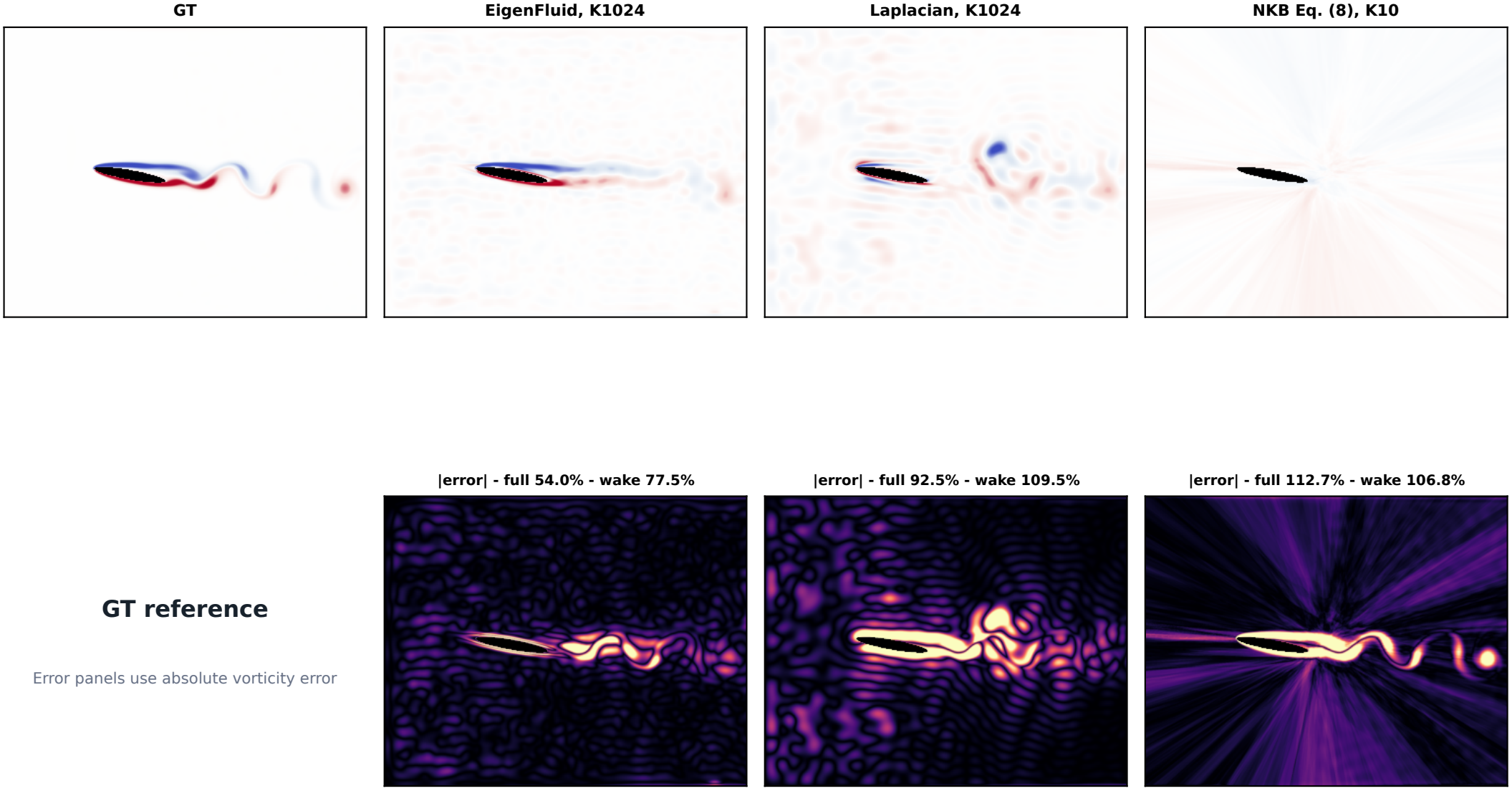
Vorticity rollout - frame 10/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



Vorticity rollout - frame 11/30

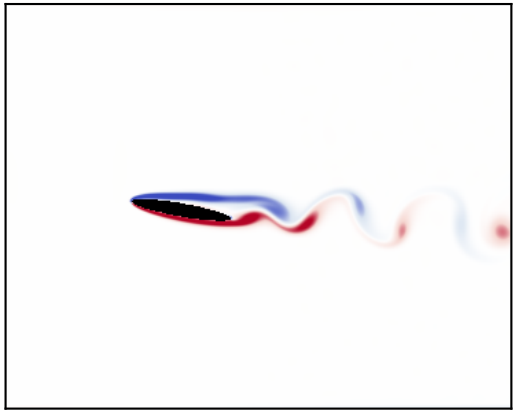
shared field scale +/-0.474; shared absolute-error scale 0...0.234



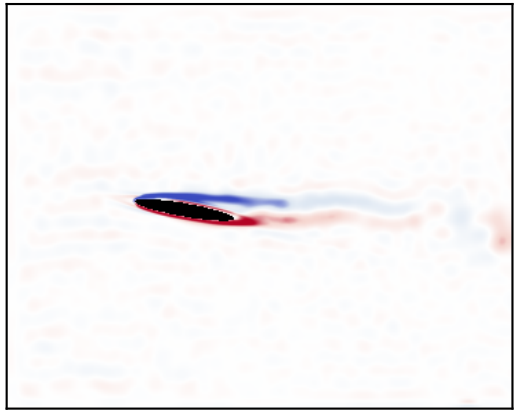
Vorticity rollout - frame 12/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234

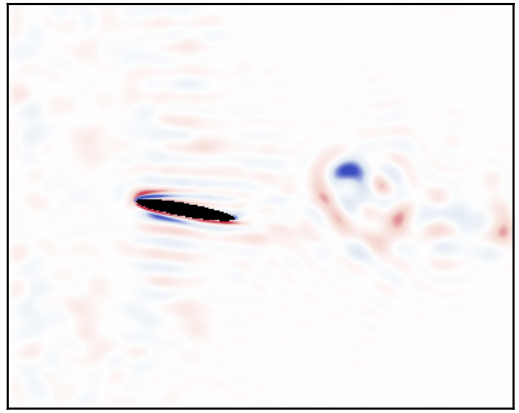
GT



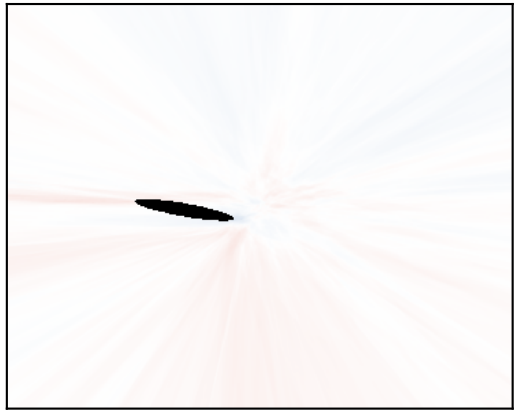
EigenFluid, K1024



Laplacian, K1024



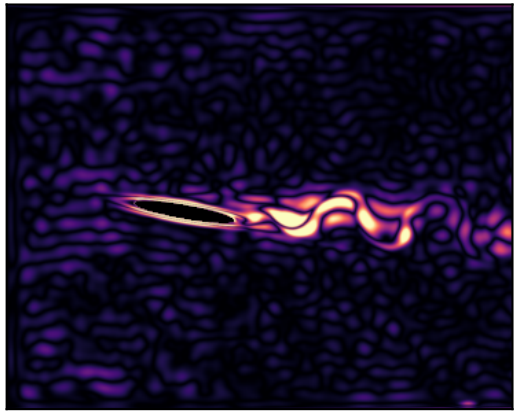
NKB Eq. (8), K10



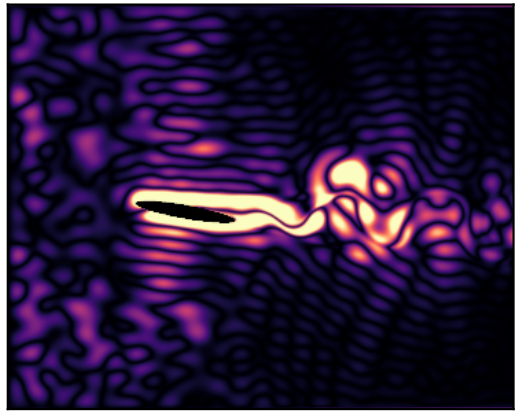
GT reference

Error panels use absolute vorticity error

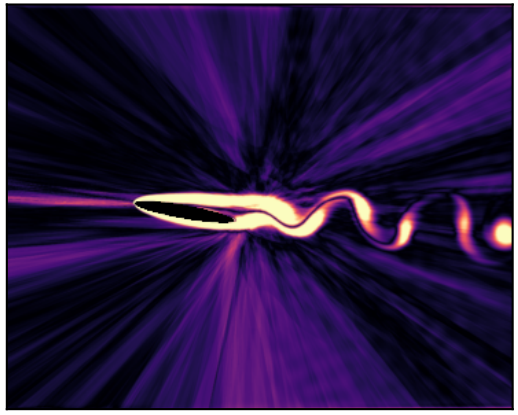
|error| - full 55.1% - wake 80.2%

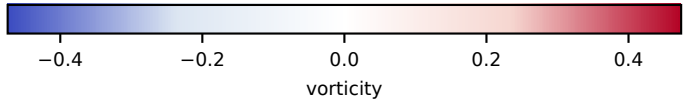


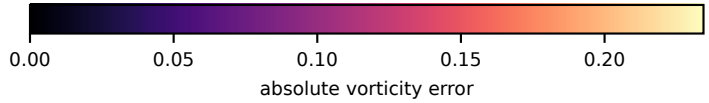
|error| - full 96.1% - wake 108.4%



|error| - full 113.0% - wake 106.8%

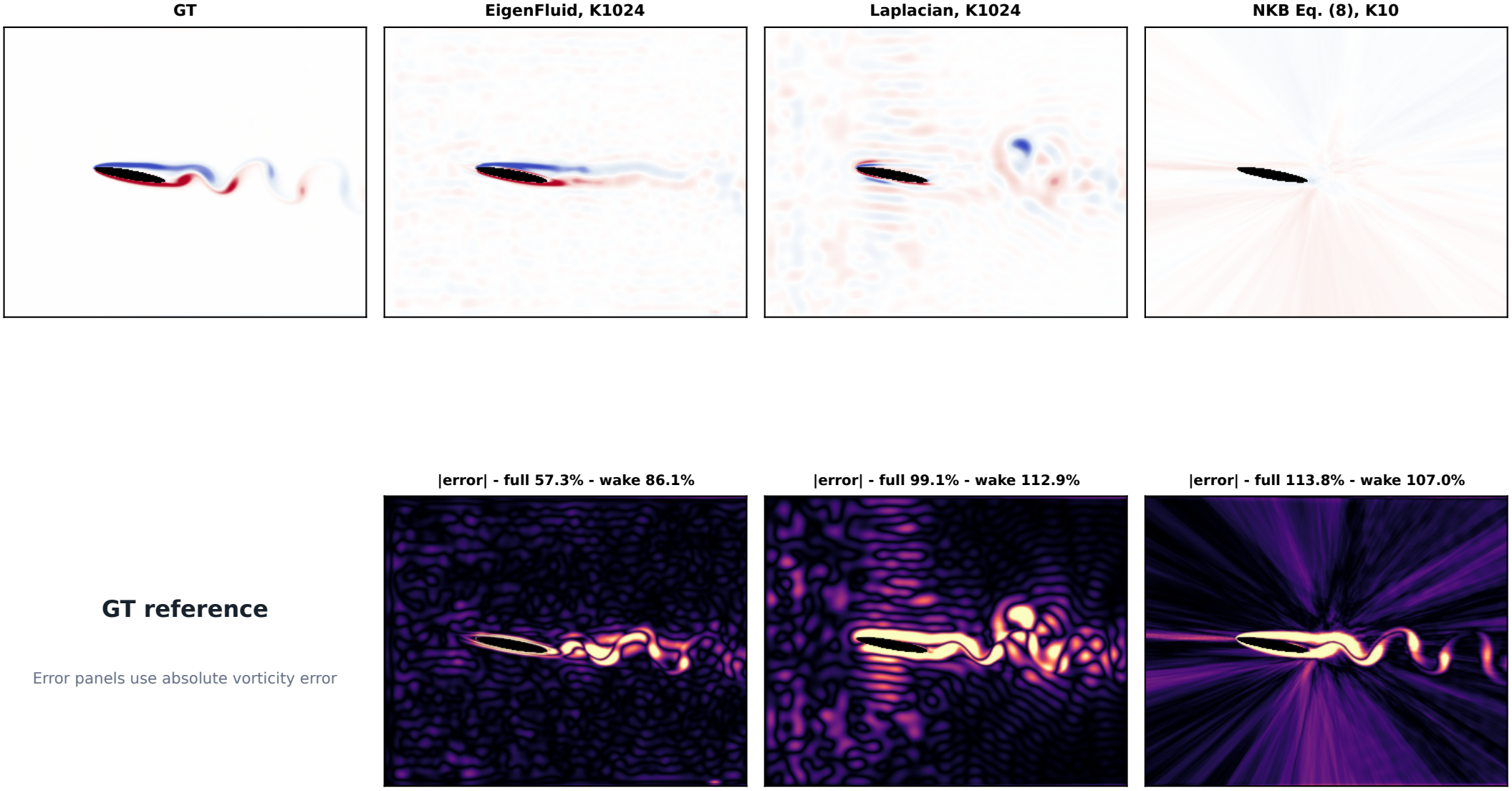






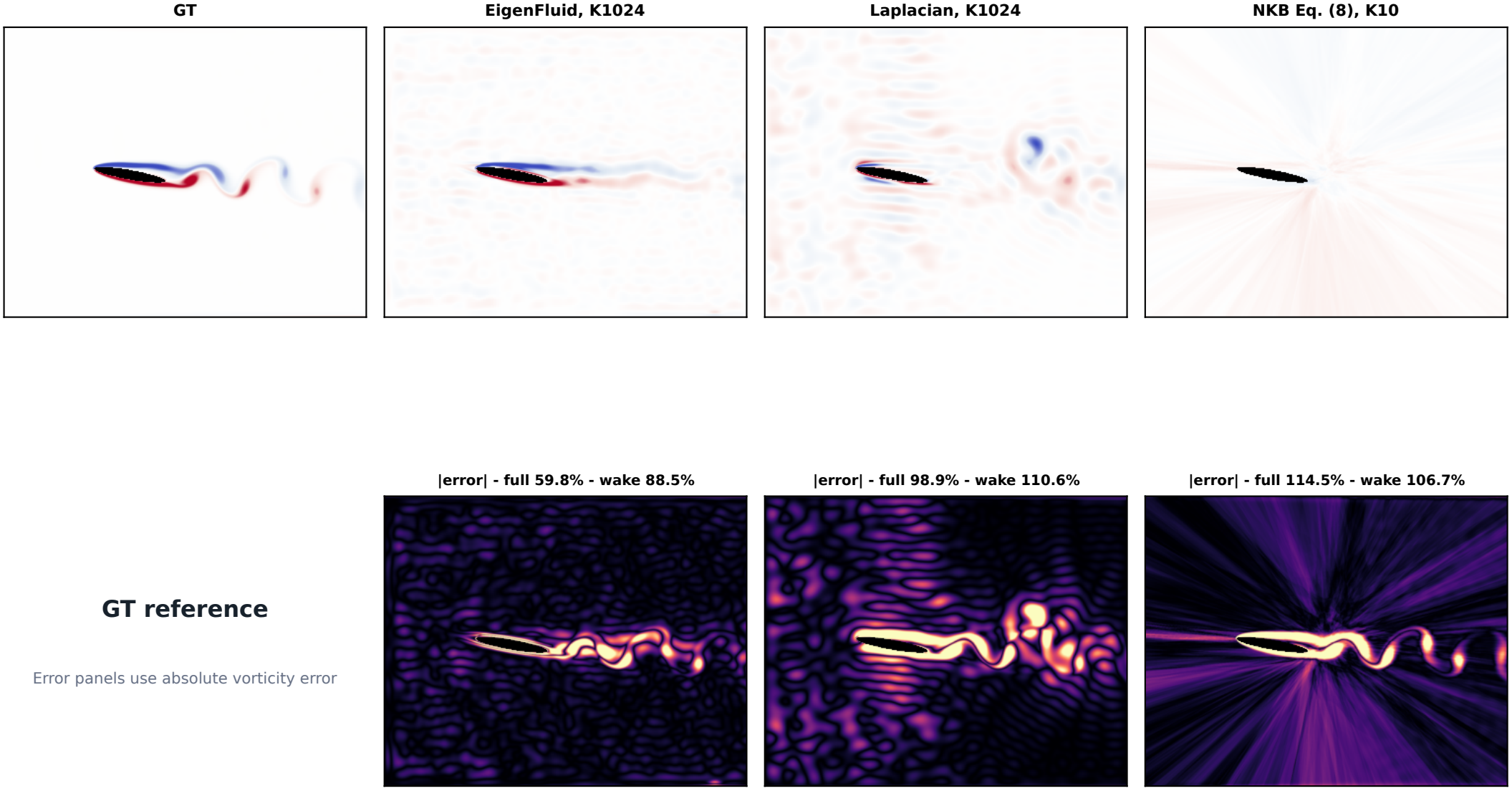
Vorticity rollout - frame 13/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



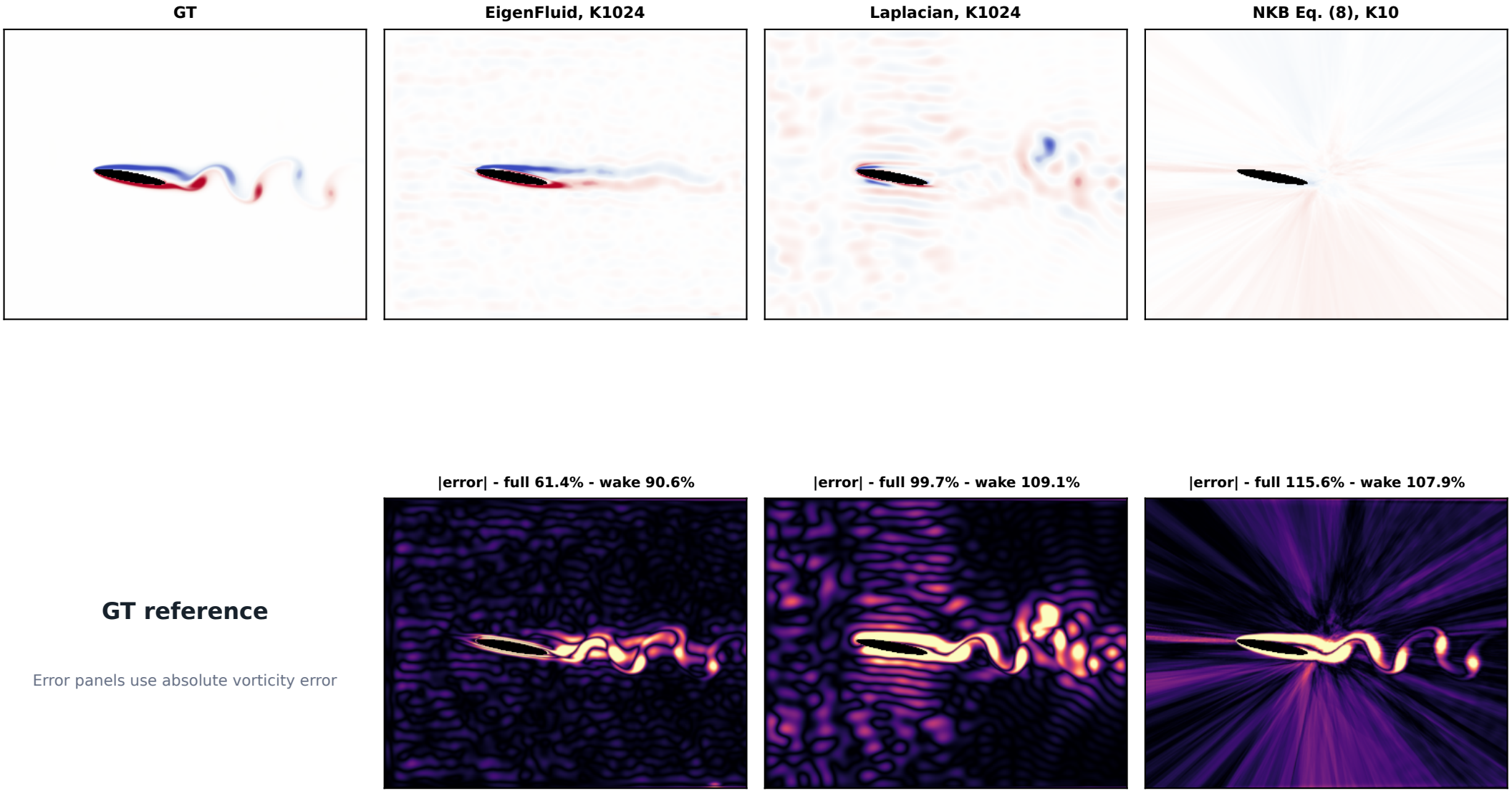
Vorticity rollout - frame 14/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



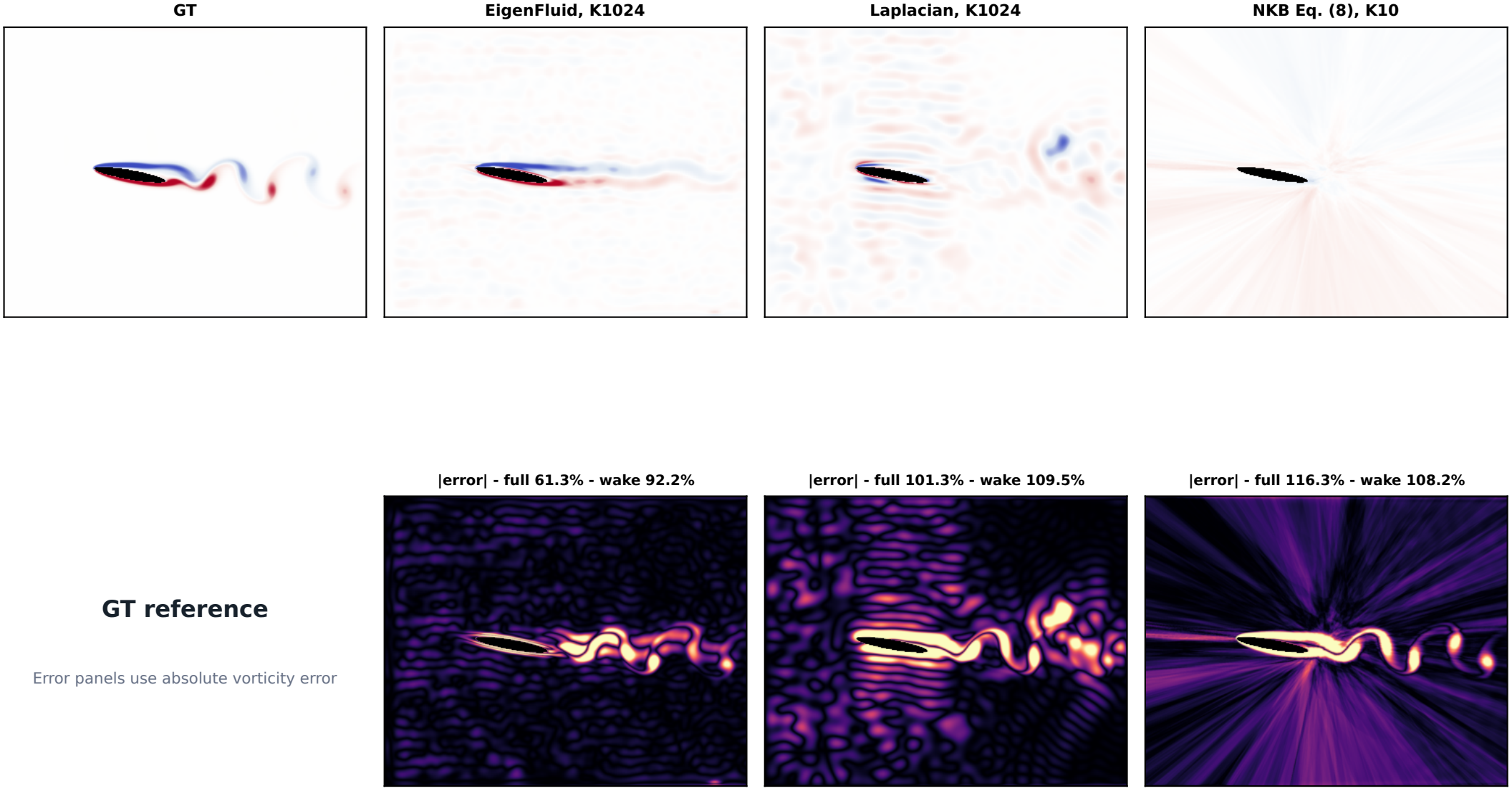
Vorticity rollout - frame 15/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



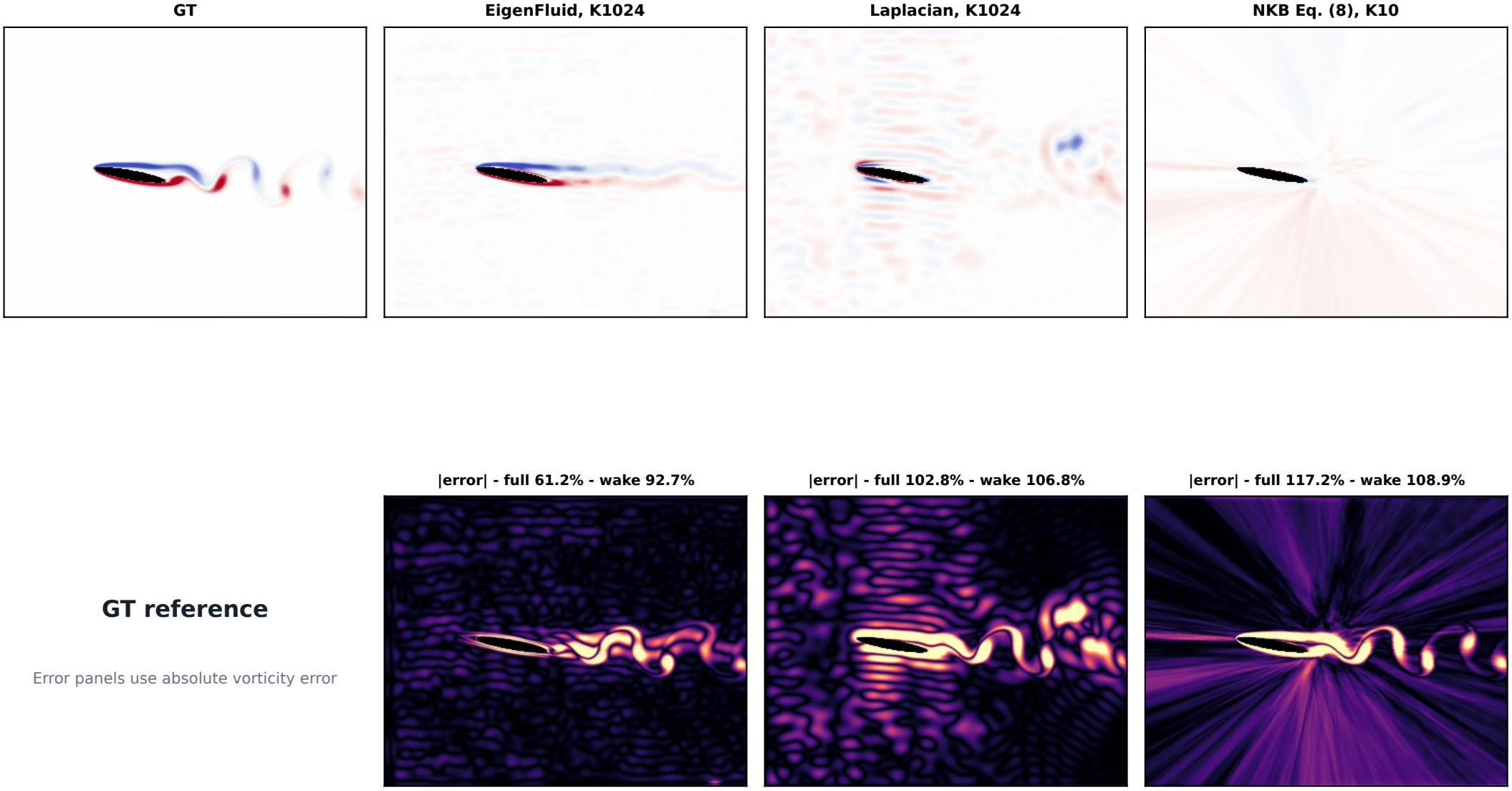
Vorticity rollout - frame 16/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



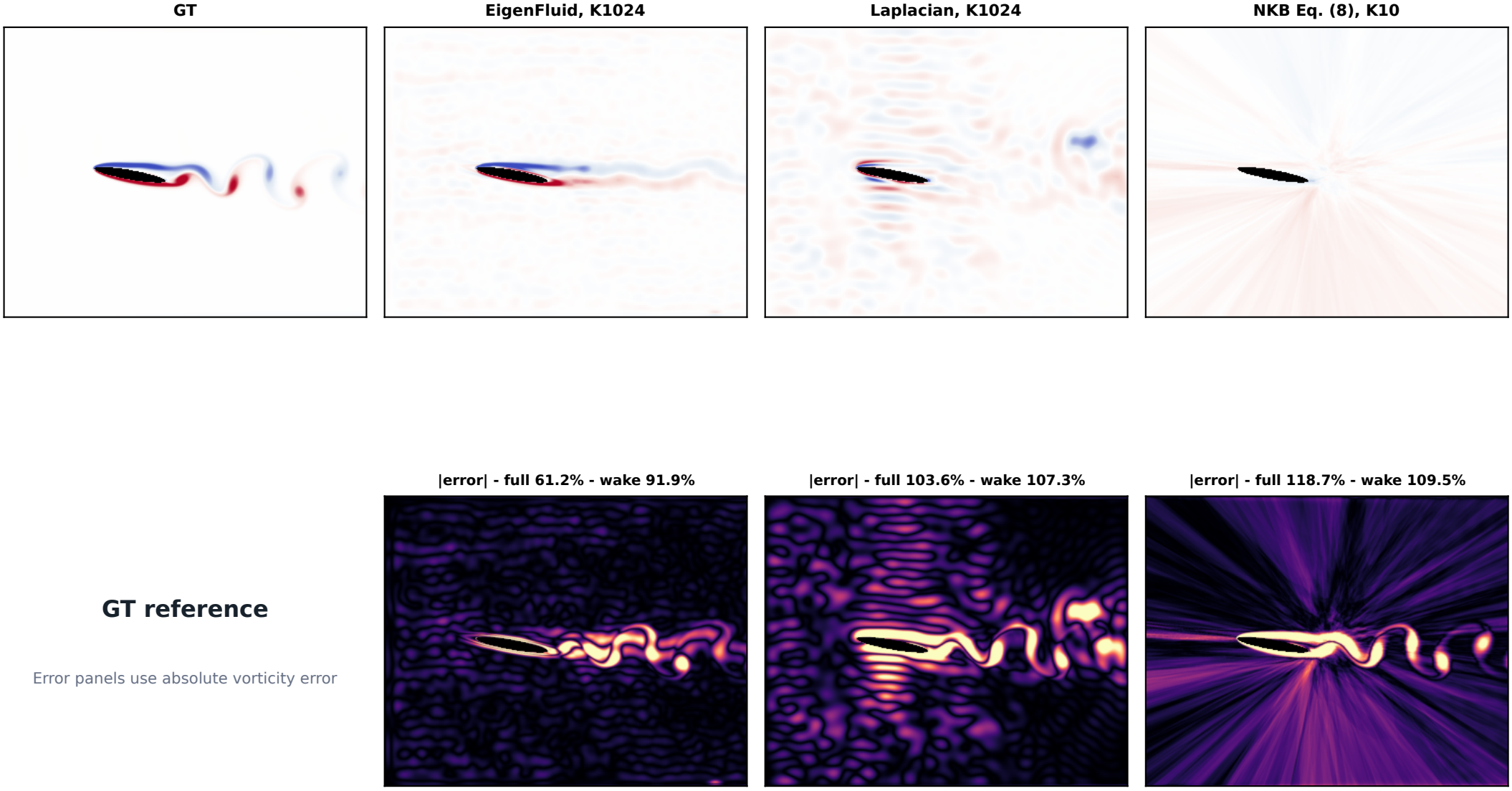
Vorticity rollout - frame 17/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



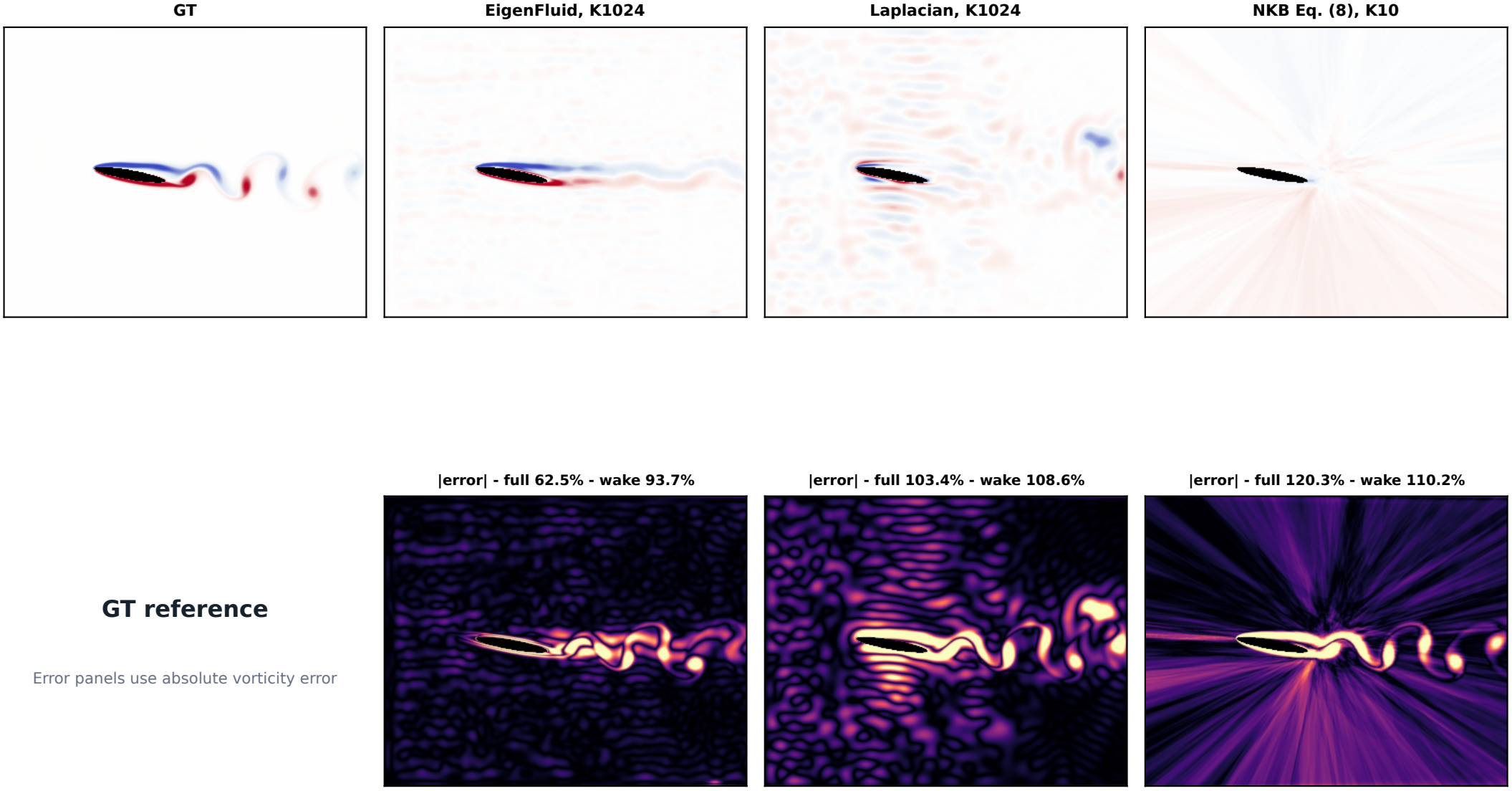
Vorticity rollout - frame 18/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



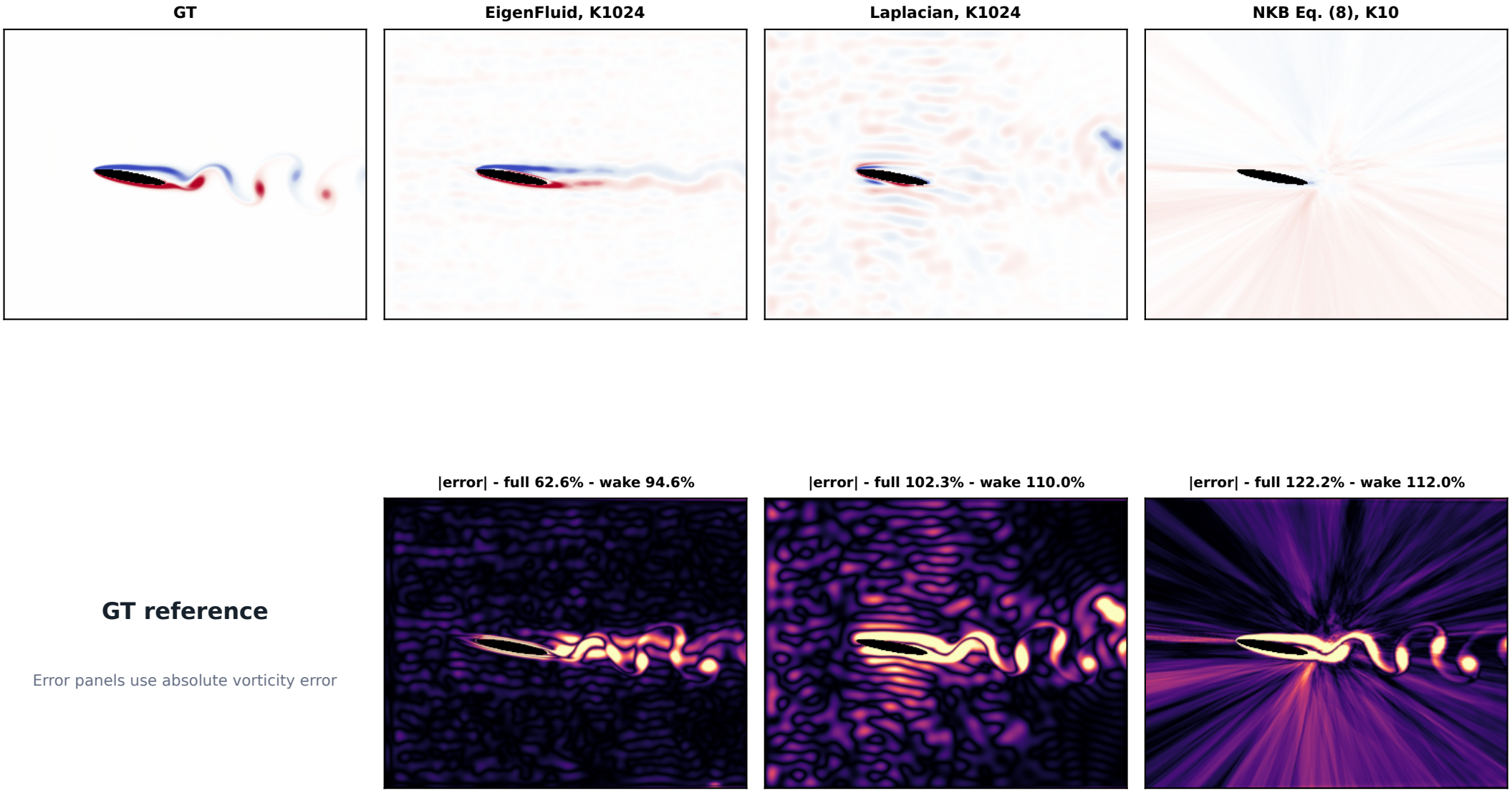
Vorticity rollout - frame 19/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



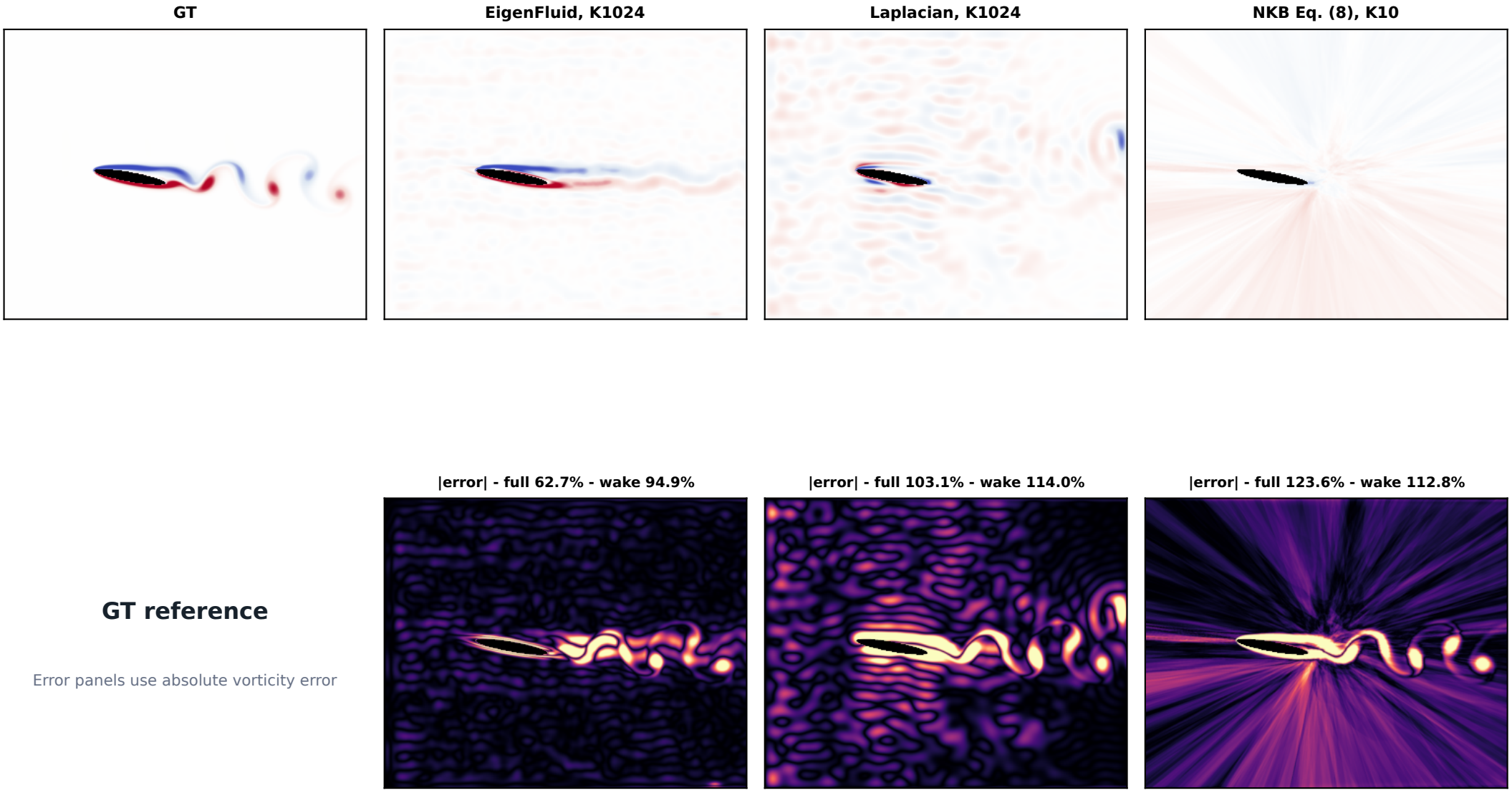
Vorticity rollout - frame 20/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



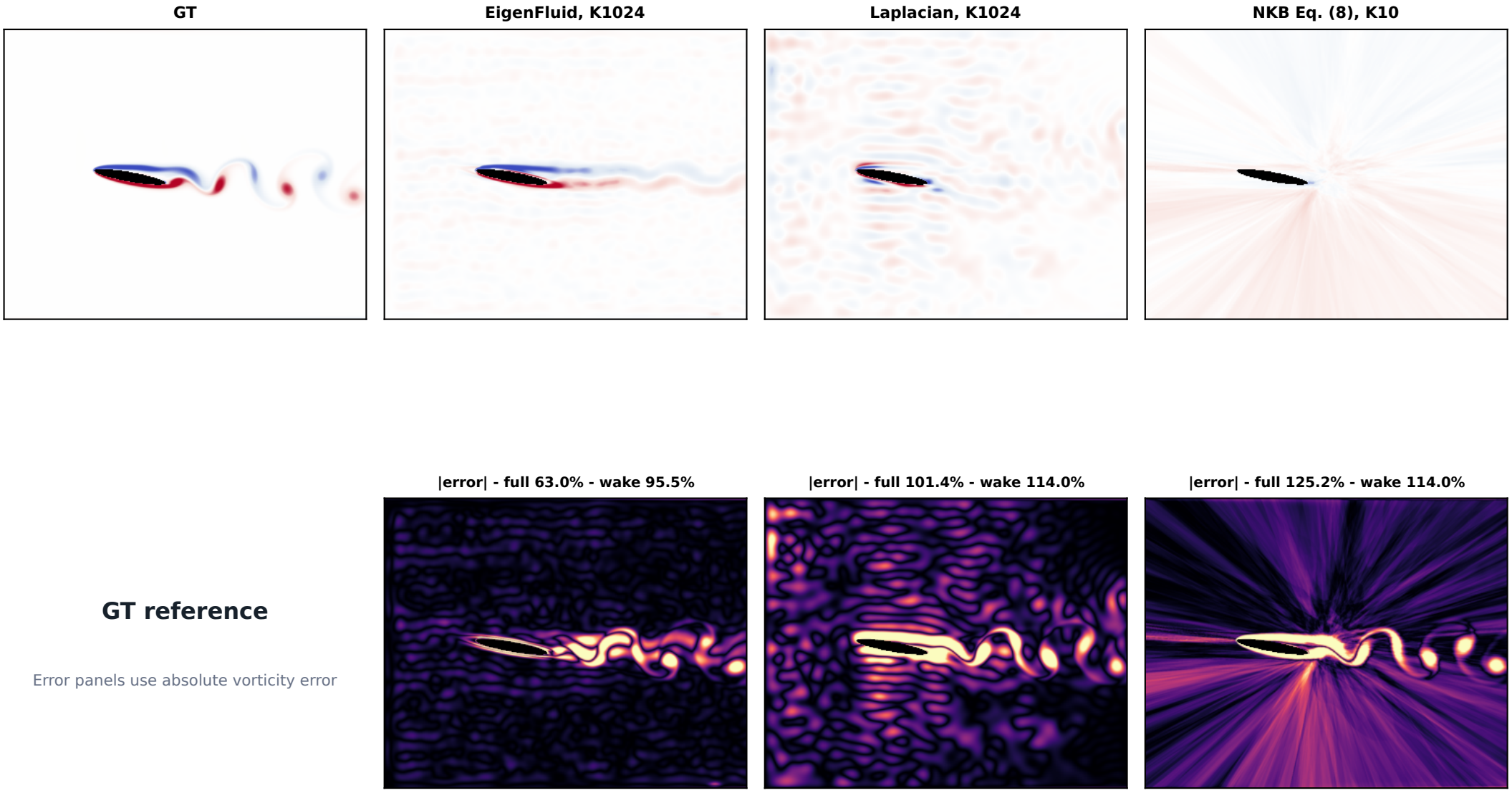
Vorticity rollout - frame 21/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



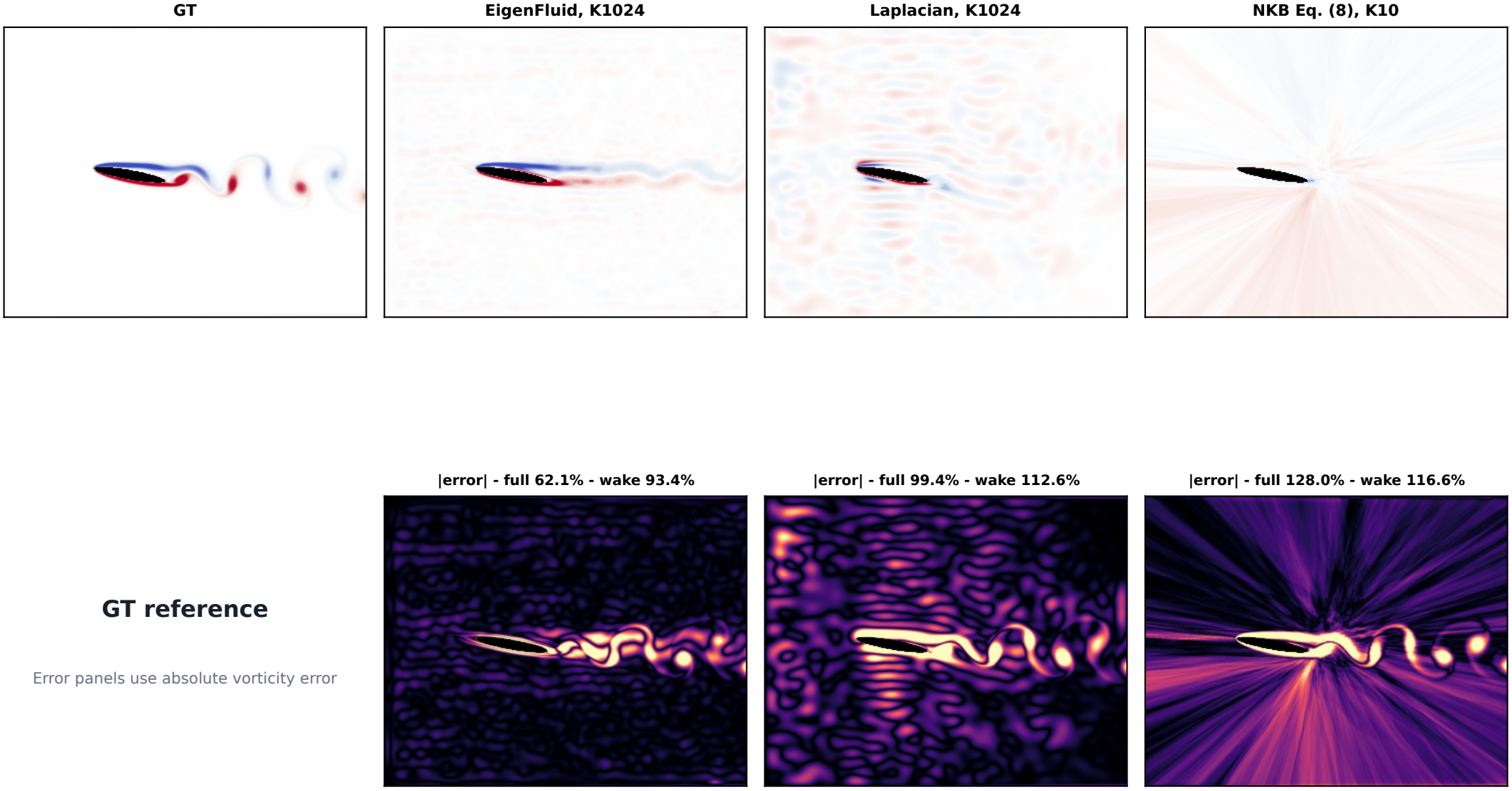
Vorticity rollout - frame 22/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



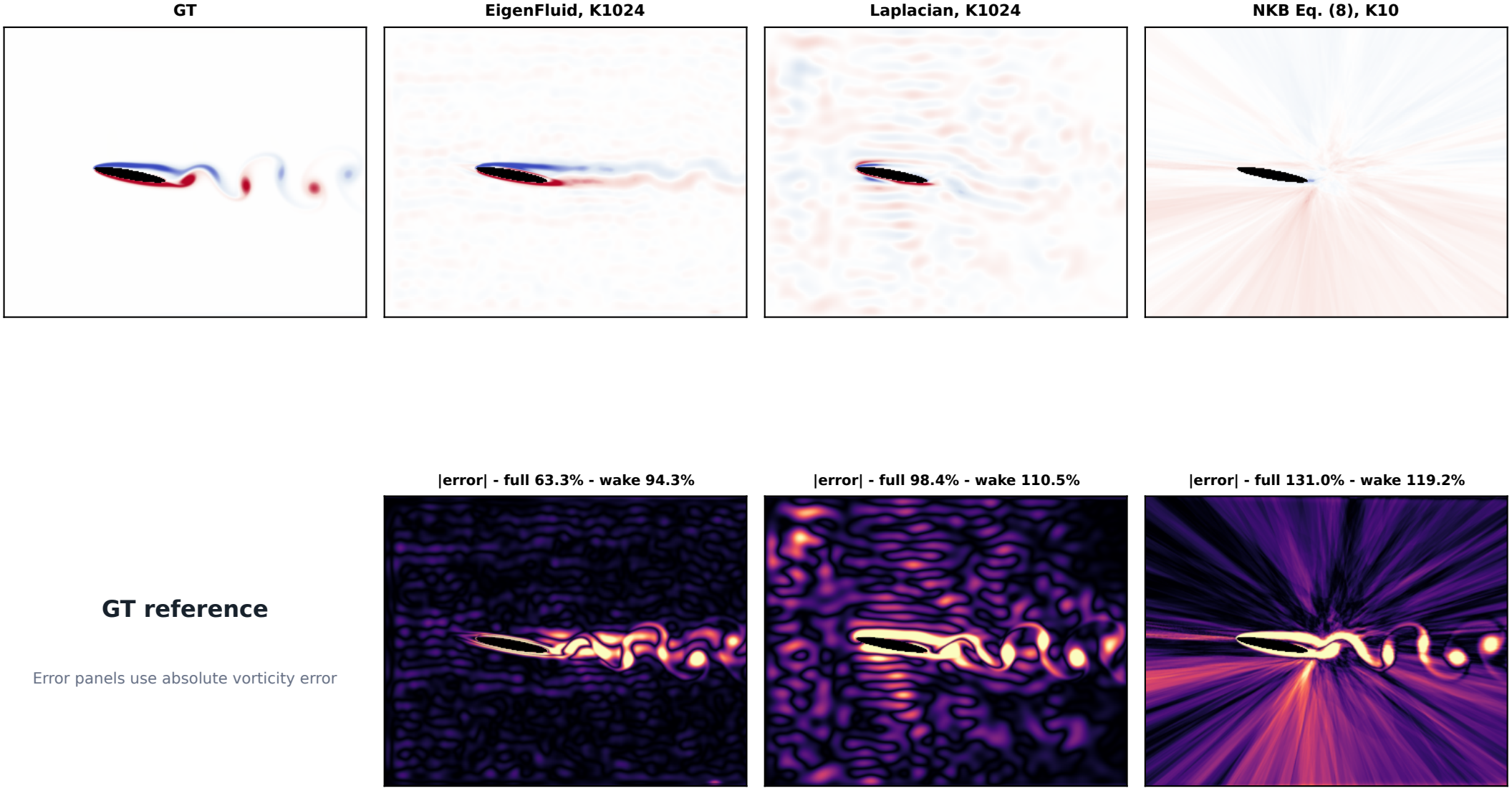
Vorticity rollout - frame 23/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



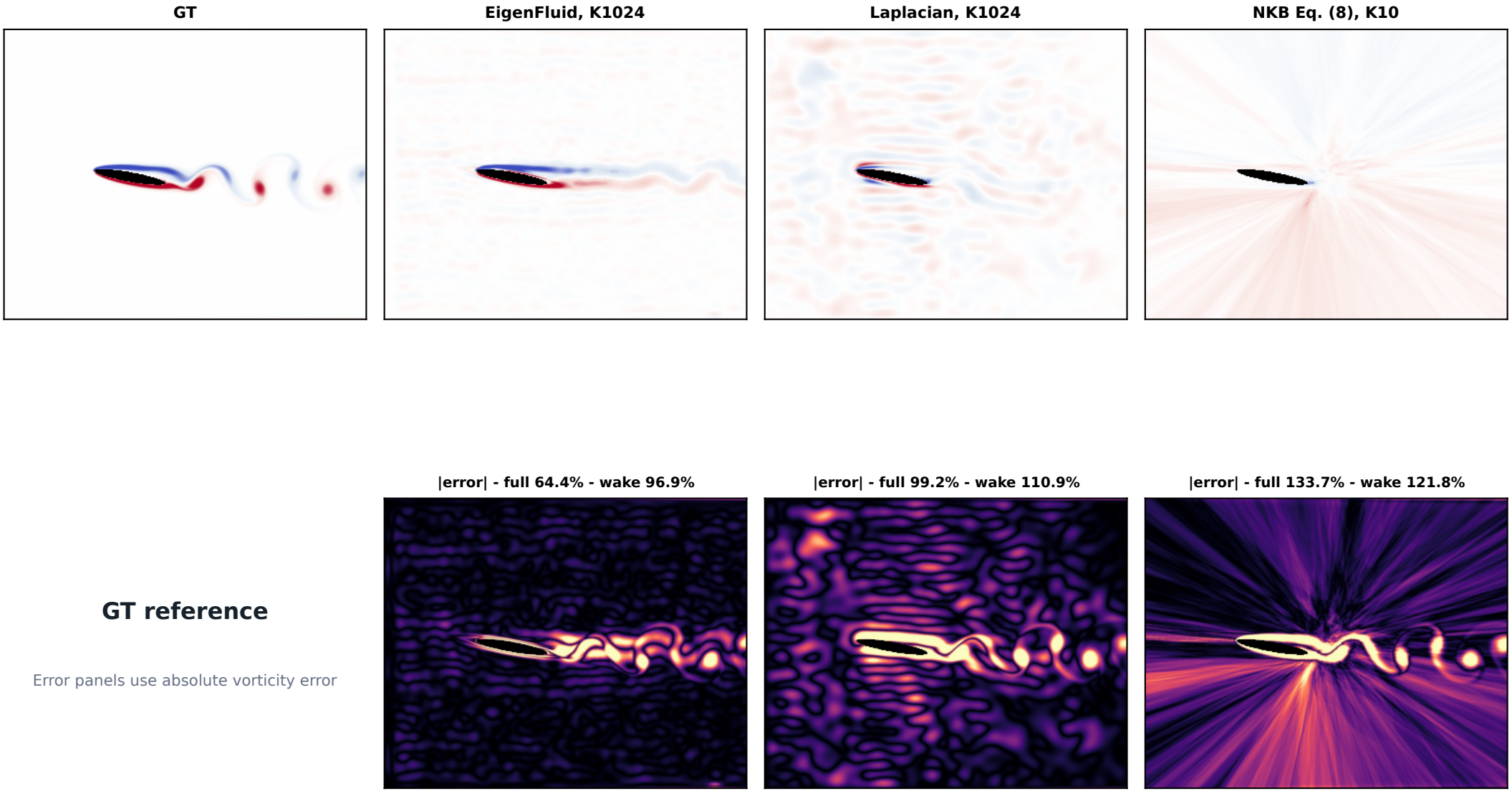
Vorticity rollout - frame 24/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



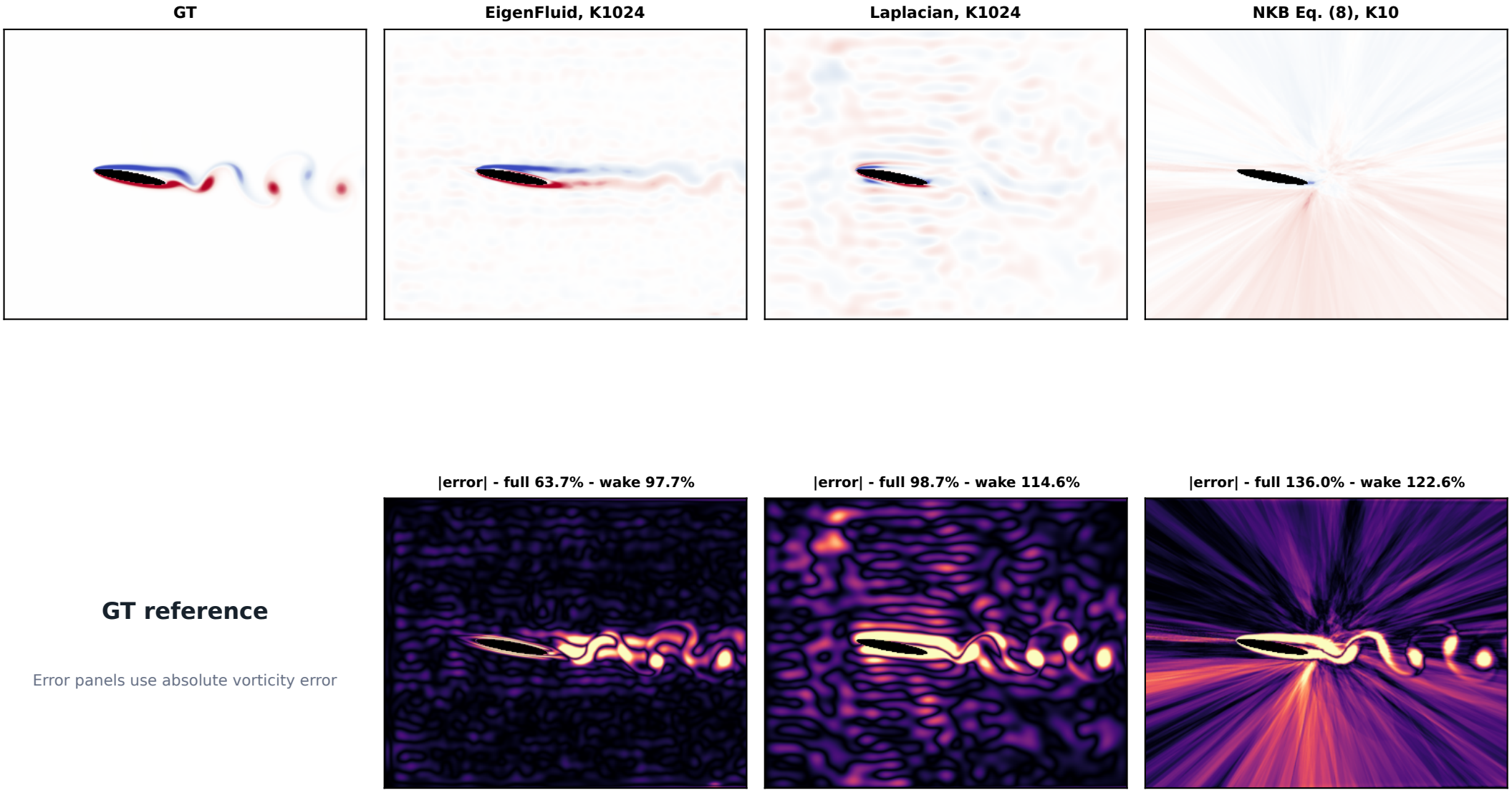
Vorticity rollout - frame 25/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



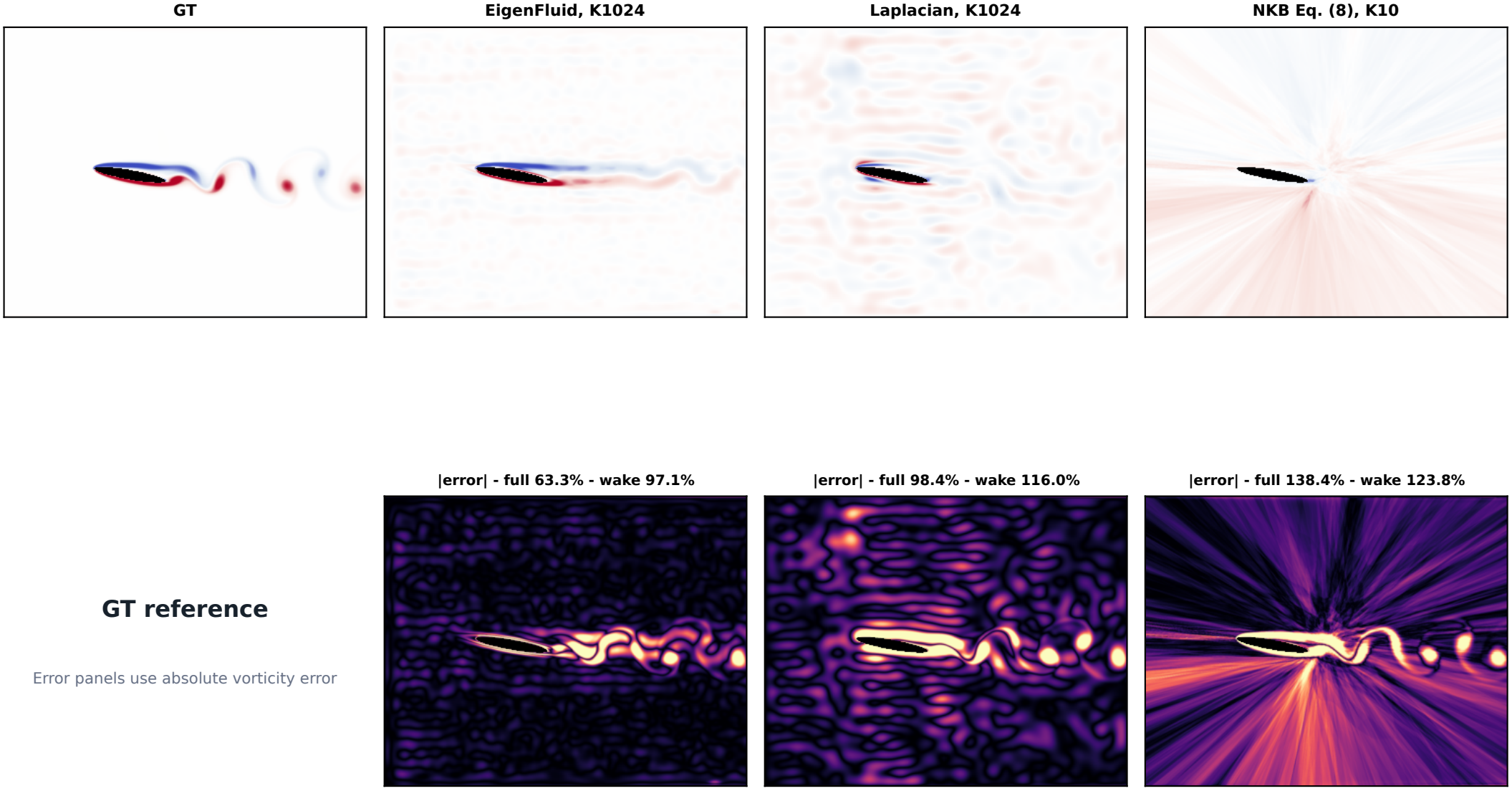
Vorticity rollout - frame 26/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



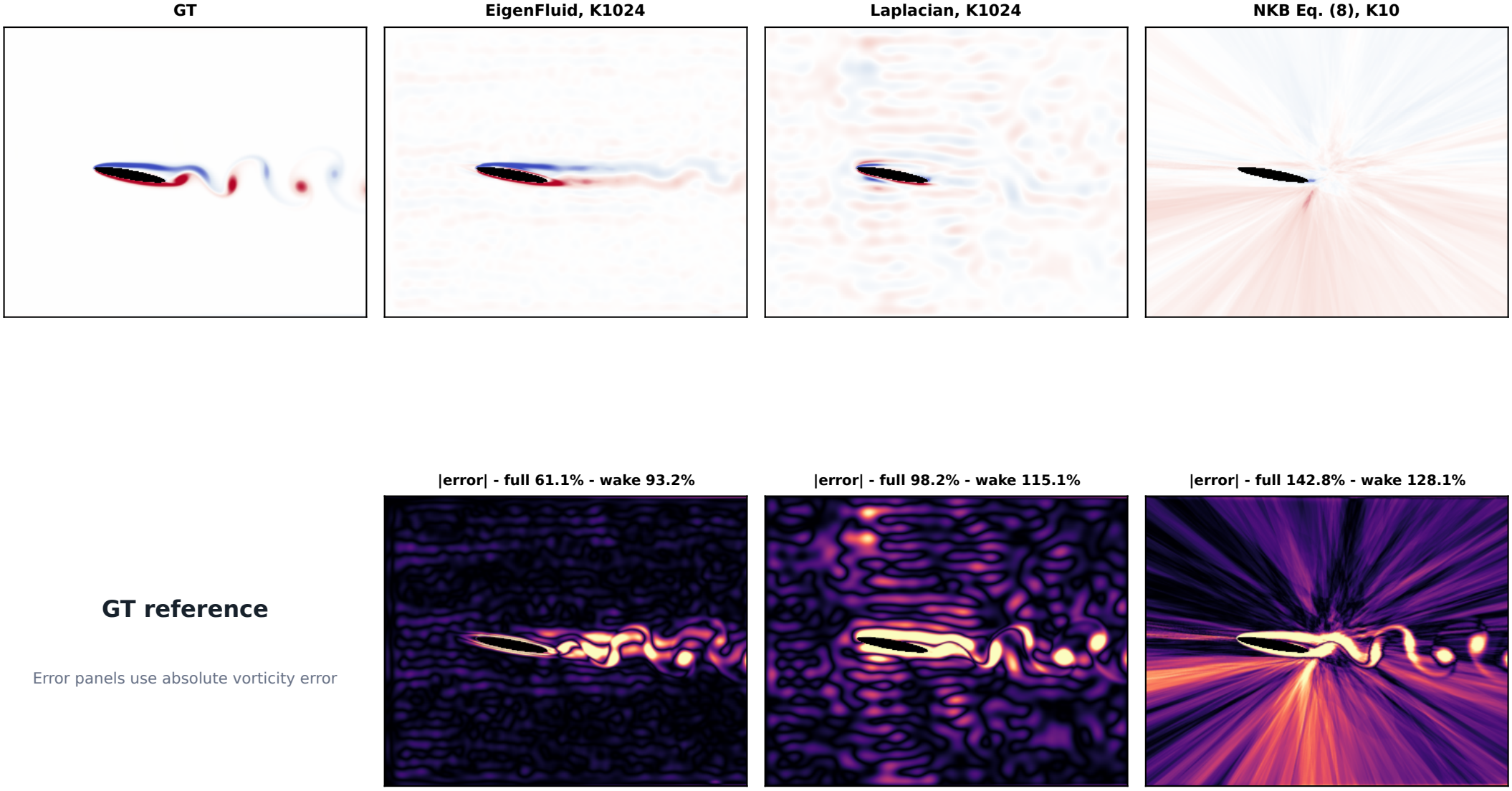
Vorticity rollout - frame 27/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



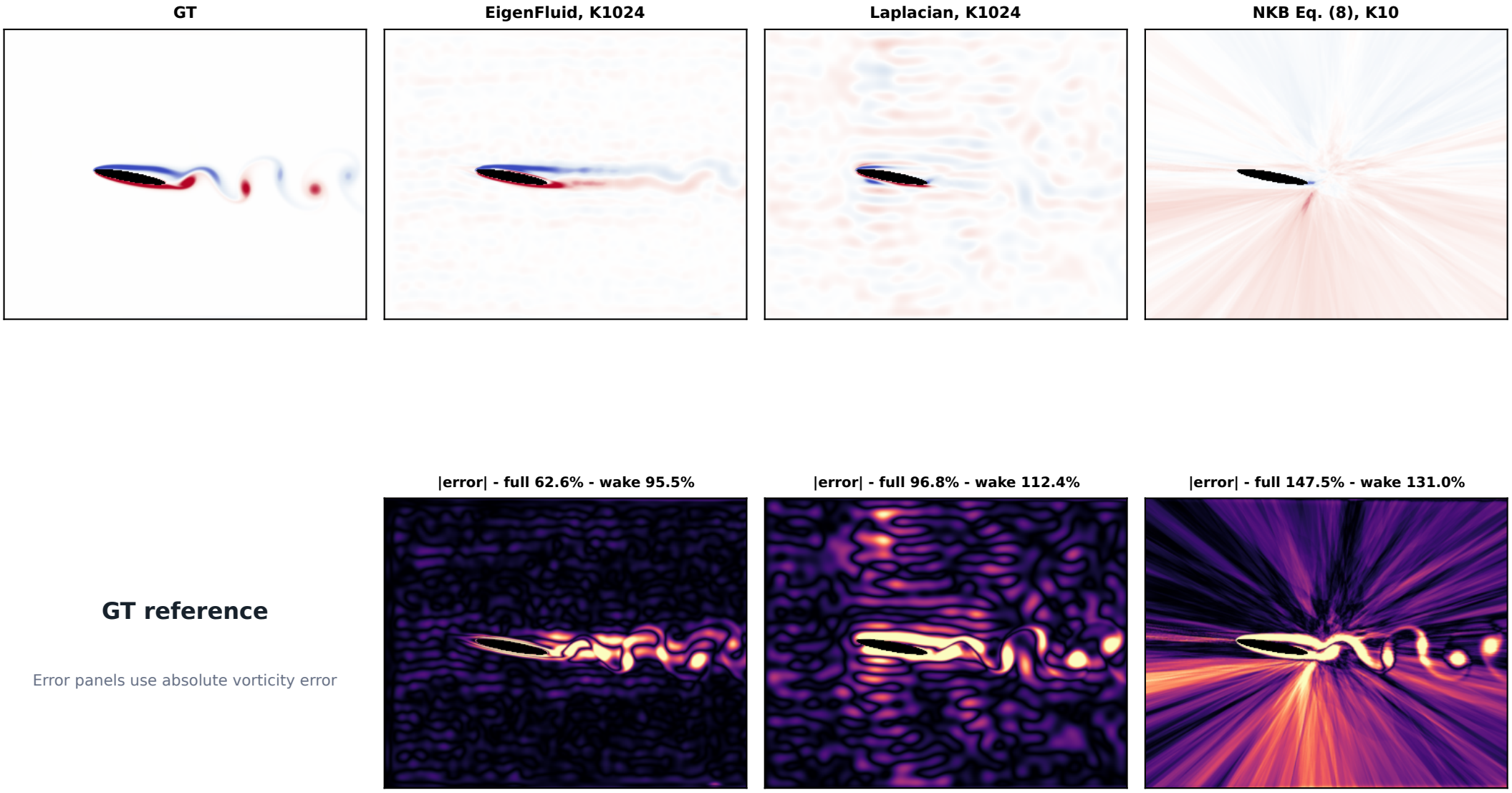
Vorticity rollout - frame 28/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



Vorticity rollout - frame 29/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234



Vorticity rollout - frame 30/30

shared field scale +/-0.474; shared absolute-error scale 0...0.234

